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Liquid ejecting apparatus

Abstract

A liquid ejecting apparatus includes a liquid ejection head, a circulation control section that controls a circulation operation of circulating the liquid in the first individual flow path, and a minute vibration control section that supplies a drive signal having a first waveform to the first piezoelectric element so as to control a minute vibration operation of causing the liquid in the first nozzle to vibrate to such a degree that the liquid is not ejected from the first nozzle. The circulation control section starts the circulation operation at a first time, and the minute vibration control section starts the minute vibration operation at a second time later than the first time.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-074876, filed Apr. 28, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a liquid ejecting apparatus.

2. Related Art

(3) In a liquid ejecting apparatus that ejects liquid such as ink, for example, thickening of liquid due to evaporation of a solvent such as moisture contained in the liquid causes a problem. The thickening of the liquid causes a decrease in ejection property, that is, a decrease in one or both of ejection amount and ejection rate. Disclosed in JP-A-2013-163290 is a liquid ejecting apparatus including a nozzle that ejects liquid and an individual flow path through which liquid is supplied to the nozzle and through which liquid not ejected from the nozzle is discharged. The liquid ejecting

apparatus performs a circulation operation of circulating liquid in the individual flow path and a minute vibration operation of causing minute vibration of liquid, of which thickening has been progressed, so that local thickening in the vicinity of the nozzle is eliminated.

(4) However, in the above-described liquid ejecting apparatus in the related art, a decrease in ejection property caused by thickening occurs in some cases even when the circulation operation and the minute vibration operation are performed.

SUMMARY

(5) According to an aspect of the present disclosure, there is provided a liquid ejecting apparatus including a liquid ejection head including a first piezoelectric element that is driven in response to supply of a drive signal, a first nozzle that ejects liquid by means of a pressure that is applied when the first piezoelectric element is driven, and a first individual flow path that communicates with the first nozzle, through which the liquid is supplied to the first nozzle, and through which the liquid not ejected from the first nozzle is discharged, a circulation control section that controls a circulation operation of circulating the liquid in the first individual flow path, and a minute vibration control section that supplies a drive signal having a first waveform to the first piezoelectric element so as to control a minute vibration operation of causing the liquid in the first nozzle to vibrate to such a degree that the liquid is not ejected from the first nozzle. The circulation control section starts the circulation operation at a first time, and the minute vibration control section starts the minute vibration operation at a second time later than the first time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a functional block diagram showing an example of the configuration of a liquid ejecting apparatus according to a first embodiment.

(2) FIG. 2 is a schematic diagram showing the liquid ejecting apparatus.

(3) FIG. 3 is a perspective view of a liquid ejection module.

(4) FIG. 4 is an exploded perspective view of a liquid ejection head shown in FIG. 3.

(5) FIG. 5 is a plan view schematically showing flow paths of a head main body.

(6) FIG. 6 is a cross-sectional view of the head main body.

(7) FIG. 7 is an enlarged view of the vicinities of piezoelectric elements of FIG. 6.

(8) FIG. 8 is a plan view of a holder.

(9) FIG. 9 is a perspective view showing flow paths provided in the holder and the head main bodies.

(10) FIG. 10 is a sectional view taken along line X-X in FIG. 8.

(11) FIG. 11 is a block diagram showing an example of the configuration of the head main body.

(12) FIG. 12 is a view showing how the vicinity of a nozzle is before a minute vibration operation is started in a comparative embodiment.

(13) FIG. 13 is a view showing how the vicinity of the nozzle is after the minute vibration operation is started in the comparative embodiment.

(14) FIG. 14 is a view showing how the vicinity of the nozzle is immediately after a circulation operation is started in the comparative embodiment.

(15) FIG. 15 is a view showing how the vicinity of the nozzle is after the circulation operation is performed for a certain period in the comparative embodiment.

(16) FIG. 16 is a diagram showing a series of operations in the liquid ejecting apparatus.

(17) FIG. 17 is a view showing how the vicinity of the nozzle is at a certain time.

(18) FIG. 18 is a view showing how the vicinity of the nozzle is in a certain period.

(19) FIG. 19 is a view showing how the vicinity of the nozzle is at a certain time.

(20) FIG. 20 is a view showing how the vicinity of the nozzle is in a certain period.

(21) FIG. **21** is a timing chart for description of the operation of the liquid ejecting apparatus in a certain period.

(22) FIG. **22** is a description diagram for description of generation of coupling state designation signals.

(23) FIG. **23** is a functional block diagram showing an example of the configuration of a liquid ejecting apparatus according to a second embodiment.

(24) FIG. **24** is a diagram showing a series of operations of the liquid ejecting apparatus.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(25) Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. Note that, the dimensions and scales of each part shown in each drawing are different from the actual dimensions and scales as appropriate. In addition, since the embodiments to be described below are suitable specific examples of the present disclosure, various technically-preferable limitations are provided to the embodiments. However, the range of the present disclosure is not limited to the embodiments unless there is no description to the effect that the present disclosure is limited hereinafter.

(26) For the sake of convenience, the following description will be made while appropriately using an X-axis, a Y-axis, and a Z-axis that intersect with each other. In addition, a direction extending along the X-axis is an X1 direction, and a direction opposite to the X1 direction is an X2 direction. Similarly, directions that extend along the Y-axis and are opposite to each other are a Y1 direction and a Y2 direction. In addition, directions that extend along the Z-axis and are opposite to each other are a Z1 direction and a Z2 direction.

(27) Here, typically, the Z-axis is a vertical axis, and the Z2 direction corresponds to a downward vertical direction. However, the Z-axis does not need to be the vertical axis and may be inclined with respect to the vertical axis. In addition, although the X-axis, the Y-axis, and the Z-axis are typically orthogonal to each other, the present disclosure is not limited thereto. For example, the X-axis, the Y-axis, and the Z-axis only need to intersect each other at an angle within a range of 80 degrees or more and 100 degrees or less.

1. First Embodiment

(28) 1-1. Liquid Ejecting Apparatus **100**

(29) The configuration of a liquid ejecting apparatus **100** will be described with reference to FIGS. **1** and **2**. FIG. **1** is a functional block diagram showing an example of the configuration of the liquid ejecting apparatus **100** according to a first embodiment. FIG. **2** is a schematic diagram showing the liquid ejecting apparatus **100**. The liquid ejecting apparatus **100** is an ink jet printing apparatus that ejects ink, which is an example of liquid, onto a medium PP in the form of droplets. The liquid ejecting apparatus **100** of the first embodiment is a so-called line-type printing apparatus in which a plurality of nozzles N for ejection of ink are distributed over the entire medium PP in the width direction. The medium PP is, typically, printing paper. Note that, the medium PP is not limited to printing paper and may be a printing target formed of any material such as a resin film or cloth.

(30) The ink according to the first embodiment has a feature that thickening of the ink is easily progressed since a solvent such as moisture contained in the ink is volatile and is easily evaporated or a feature that thickening of the ink is easily progressed when a solvent is evaporated.

(31) As shown in FIGS. **1** and **2**, the liquid ejecting apparatus **100** includes a liquid supply source **110**, a control module **120**, a transport mechanism **130**, a liquid ejection module **140**, a circulation mechanism **150**, a pump **170**, a capping mechanism **180**, and a drive signal generation circuit **190**.

(32) The liquid supply source **110** is a container storing ink. Specific examples of the liquid supply source **110** include a cartridge detachable from the liquid ejecting apparatus **100**, a bag-shaped ink pack formed of a flexible film, and an ink tank refillable with ink. Note that, any type of ink is stored in the liquid supply source **110**.

(33) Although not shown in the drawings, the liquid supply source **110** of the first embodiment includes a first liquid container and a second liquid container. First ink is stored in the first liquid

container. Second ink, which is a different type of ink from the first ink, is stored in the second liquid container. For example, the first ink and the second ink are inks of which the colors are different from each other. The first ink and the second ink may be the same type of ink.

(34) The control module **120** controls the operation of each element in the liquid ejecting apparatus **100**. The control module **120** includes, for example, one or more processing circuits such as CPUs or FPGAs, and one or more storage circuits such as semiconductor memories. “CPU” is an abbreviation for “Central Processing Unit”. “FPGA” is an abbreviation for “Field Programmable Gate Array”. Various programs and various data are stored in the storage circuit. The processing circuit realizes various types of control by executing the programs and appropriately using the data.

(35) The transport mechanism **130** transports the medium PP in a direction DM under the control of the control module **120**. The direction DM in the first embodiment is the Y2 direction. In an example shown in FIG. 2, the transport mechanism **130** includes a transport roller that is long along the X-axis and a motor that rotates the transport roller. Note that, the configuration of the transport mechanism **130** is not limited to a configuration in which the transport roller is used, and may be a configuration in which a drum or an endless belt that transports the medium PP sticking fast to an outer peripheral surface because of an electrostatic force or the like, for example.

(36) Under the control of the control module **120**, the liquid ejection module **140** ejects ink to the medium PP in the Z2 direction from each of the plurality of nozzles N, the ink being supplied from the liquid supply source **110** via the circulation mechanism **150**. The liquid ejection module **140** is a line head including a plurality of liquid ejection heads **10** that are disposed such that the plurality of nozzles N are distributed over the entire medium PP in a direction along the X-axis. That is, a group of the plurality of liquid ejection heads **10** constitutes a long line head that extends in the direction along the X-axis. When ejection of ink from the plurality of liquid ejection heads **10** is performed in parallel with transportation of the medium PP which is performed by the transport mechanism **130**, an image is formed on a surface of the medium PP by means of ink. Note that, the plurality of nozzles N that one liquid ejection head **10** includes may be disposed to be distributed over the entire medium PP in the direction along the X-axis and in this case, for example, the liquid ejection module **140** is composed of the one liquid ejection head **10**.

(37) The liquid supply source **110** is coupled to the liquid ejection module **140** via the circulation mechanism **150**. The circulation mechanism **150** is a mechanism that supplies ink to the liquid ejection module **140** and that retrieves ink discharged from the liquid ejection module **140** for re-supply to the liquid ejection module **140** under the control of the control module **120**. In an example shown in FIG. 2, the circulation mechanism **150** includes a sub tank **151** for storage of ink, a supply flow path **153** for supply of ink from the sub tank **151** to the liquid ejection module **140**, a retrieval flow path **155** for retrieval of ink from the liquid ejection module **140** to the sub tank **151**, and a pump **157** that causes ink to appropriately flow. The sub tank **151**, the supply flow path **153**, the retrieval flow path **155**, and the pump **157** are provided for each of the above-described first liquid container and the above-described second liquid container. Ink is supplied from the liquid supply source **110** to the sub tank **151** with the control module **120** controlling the pump **170**. With the circulation mechanism **150** being operated, it is possible to suppress an increase in viscosity of ink and to suppress retention of air bubbles in ink.

(38) The supply flow path **153** is positioned upstream of the liquid ejection module **140** in a direction in which ink flows. The retrieval flow path **155** is positioned downstream of the liquid ejection module **140** in the direction in which ink flows. In the example of FIG. 2, the pump **157** is provided in the supply flow path **153**. That is, the pump **157** is provided upstream of the liquid ejection module **140**. However, the pump **157** may be provided in the retrieval flow path **155** (that is, the pump **157** may be provided downstream of the liquid ejection module **140**). Alternatively, the circulation mechanism **150** may include a plurality of pumps, a portion of the plurality of pumps may be provided in the supply flow path **153**, and the remainder of the plurality of pumps may be provided in the retrieval flow path **155**. In addition, the circulation mechanism **150** may

include a compressor instead of the pump **157**. The pump **157** and the compressor are an example of “one or more flow mechanisms provided at one or both of a position upstream of a liquid ejection head and a position downstream of the liquid ejection head”.

(39) The capping mechanism **180** is a mechanism provided to seal nozzle surfaces FN provided with the nozzles N. The capping mechanism **180** includes a cap **182** and a cap mover **184**. The cap **182** seals the nozzle surfaces FN provided with the nozzles N. The cap mover **184** moves the cap **182** relative to the liquid ejection module **140** along the Y-axis and the Z-axis under the control of the control module **120**. The cap mover **184** is composed of, for example, a guide rail, a motor, and the like. The cap mover **184** moves the cap **182** relative to each of the plurality of liquid ejection heads **10** in the Z1 direction during a period in which ink is not ejected to the medium PP from each of the plurality of liquid ejection heads **10** so that a tip end portion of the cap **182** comes into contact with the nozzle surfaces FN of the plurality of liquid ejection heads **10** and the nozzle surfaces FN are covered by the cap **182** at least partially. The nozzle surfaces FN will be described later with reference to FIG. 6.

(40) Hereinafter, an operation of sealing the nozzle surfaces FN by means of the cap **182** will be described as a “cap sealing operation”, and an operation of unsealing the nozzle surfaces FN sealed by the cap **182** will be described as “cap unsealing operation”. Note that, the cap mover **184** may cause the nozzle surfaces FN to be sealed by moving the liquid ejection module **140** instead of moving the cap **182**.

(41) The liquid ejection head **10** includes one or more head main bodies **14** that eject ink. In the first embodiment, one liquid ejection head **10** includes six head main bodies **14**. The head main body **14** includes M ejectors D that eject ink and a switching circuit **141**. In the first embodiment, M is an integer of 2 or more. Note that M may also be 1.

(42) Hereinafter, for distinction between the M ejectors D that one head main body **14** includes, the ejectors D may be referred to as a first-stage ejector, a second-stage ejector, . . . , and an Mth-stage ejector in order. In addition, an mth-stage ejector D may be referred to as an “ejector D[m]”. In the following description, the variable m is an integer of 1 or more and M or less. In addition, when a component of the liquid ejecting apparatus **100**, a signal, or the like corresponds to the ordinal number “m” of the ejector D[m], a suffix “[m]” that indicating that the component, the signal, or the like corresponds to the ordinal number “m” may be added to a symbol representing the component, the signal, or the like.

(43) The switching circuit **141** switches, based on a designation signal SI, whether or not a drive signal Com output from the drive signal generation circuit **190** is to be supplied to each ejector D.

(44) The control module **120** reads a program from the storage circuit in the control module **120** and executes the read program to function as a drive control section **121**, a cap control section **125**, a circulation control section **127**, and a transport control section **129**. However, a device that functions as the drive control section **121**, a device that functions as the cap control section **125**, a device that functions as the circulation control section **127**, and a device that functions as the transport control section **129** may not be the same device. For example, a device that functions as the drive control section **121**, a device that functions as the cap control section **125**, a device that functions as the circulation control section **127**, and a device that functions as the transport control section **129** may be different devices.

(45) The drive control section **121** controls the operation of the ejectors D. More specifically, the drive control section **121** generates the designation signal SI for control of the ejectors D and a waveform designation signal dCom for control of the drive signal generation circuit **190**. The waveform designation signal dCom is a digital signal that defines the waveform of the drive signal Com. In addition, the drive signal Com is an analog signal used to drive the ejectors D. The drive signal generation circuit **190** includes a DA conversion circuit and generates the drive signal Com having a waveform defined by the waveform designation signal dCom. Note that, in the first embodiment, it will be assumed that the drive signal Com includes a drive signal Com-A and a

drive signal Com-B.

(46) In addition, the designation signal SI is a digital signal for designation of the type of operation of the ejectors D. Specifically, the designation signal SI designates the type of operation of the ejectors D by designating whether or not the drive signal Com is to be supplied to the ejectors D. Here, the designation of the type of operation of the ejectors D is, for example, designating whether or not the ejectors D are to be driven or designating whether or not ink is to be ejected from the ejectors D when the ejectors D are driven. When the ejectors D are to be driven to perform a printing ejection operation in which ink is ejected from the nozzles N of the ejectors D so that dots constituting an image are formed on the medium PP, the drive control section **121** functions as an ejection control section **122** that controls the ejection operation. Meanwhile, when the ejectors D are to be driven to perform a minute vibration operation in which ink in the nozzles N is caused to vibrate to such a degree that no ink is ejected from the nozzles N, the drive control section **121** functions as a minute vibration control section **123** that controls the minute vibration operation. The printing ejection operation is an example of an “ejection operation of causing liquid to be ejected from a first nozzle”.

(47) The cap control section **125** controls the capping mechanism **180**. More specifically, the cap control section **125** outputs a signal for control of the capping mechanism **180** to the capping mechanism **180** to control the cap unsealing operation.

(48) The circulation control section **127** controls the circulation mechanism **150**. More specifically, the circulation control section **127** outputs a signal for control of the circulation mechanism **150** to the circulation mechanism **150** to generate a pressure difference between a position upstream of the liquid ejection head **10** and a position downstream of the liquid ejection head **10** and to control a circulation operation of circulating ink in an individual flow path PJ which will be described later.

(49) The transport control section **129** controls the transport mechanism **130**. More specifically, the transport control section **129** outputs a signal for control of the transport mechanism **130** to the transport mechanism **130**.

(50) First, the control module **120** stores printing data Img in a storage circuit in the control module **120**, the printing data Img being supplied from a host computer. Next, the control module **120** generates, based on various types of data such as the printing data Img stored in the storage circuit, the designation signal SI, the waveform designation signal dCom, a signal for control of the transport mechanism **130**, a signal for control of the capping mechanism **180**, a signal for control of the circulation mechanism **150**, and a signal for control of the pump **170**. Then, the control module **120** controls, based on the various control signals and the various types of data stored in the storage circuit, the liquid ejection module **140** such that the ejectors D are driven while controlling the transport mechanism **130** such that the position of the medium PP relative to the liquid ejection module **140** is changed. In this manner, the control module **120** adjusts whether or not ink is ejected from the ejectors D, the amount of ejection of ink, the timing of ejection of ink, and the like and controls performance of a printing process of forming an image corresponding to the printing data Img on the medium PP.

(51) 1-2. Liquid Ejection Module **140**

(52) FIG. **3** is a perspective view of the liquid ejection module **140**. As shown in FIG. **3**, the liquid ejection module **140** includes a supporting body **41** and the plurality of liquid ejection heads **10**. The supporting body **41** is a member that supports the plurality of liquid ejection heads **10**. In an example shown in FIG. **3**, the supporting body **41** is a plate-shaped member formed of metal or the like, and is provided with an attachment hole **41a** for attachment of the plurality of liquid ejection heads **10**. The plurality of liquid ejection heads **10** are inserted into the attachment hole **41a** in a state of being arranged in the direction along the X-axis, and each liquid ejection head **10** is fixed to the supporting body **41** by being screwed or the like. In FIG. **3**, two liquid ejection heads **10** are shown as representatives. The number of liquid ejection heads **10** of the liquid ejection module **140** can be any number. In addition, the shape and the like of the supporting body **41** are not limited to

those in the example shown in FIG. 3, and any shape or the like can be adopted.

(53) 1-3. Liquid Ejection Head **10**

(54) FIG. 4 is an exploded perspective view of the liquid ejection head **10** shown in FIG. 3. As shown in FIG. 4, the liquid ejection head **10** includes a flow path structure **11**, a wiring substrate **12**, a holder **13**, six head main bodies **14_1**, **14_2**, **14_3**, **14_4**, **14_5**, and **14_6**, a fixation plate **15**, and a base **16**. The flow path structure **11**, the wiring substrate **12**, the holder **13**, the six head main bodies **14_1**, **14_2**, **14_3**, **14_4**, **14_5**, and **14_6**, the fixation plate **15**, and the base **16** are disposed in the order of the base **16**, the flow path structure **11**, the wiring substrate **12**, the holder **13**, the plurality of head main bodies **14_1**, **14_2**, **14_3**, **14_4**, **14_5**, and **14_6**, and the fixation plate **15** in the Z2 direction. Hereinafter, each part of the liquid ejection head **10** will be described in order. Note that, hereinafter, the head main bodies **14_1**, **14_2**, **14_3**, **14_4**, **14_5**, and **14_6** may be collectively referred to as the head main bodies **14**.

(55) The flow path structure **11** is a structure in which a flow path used to cause ink to flow between the circulation mechanism **150** and the plurality of head main bodies **14** is provided. As shown in FIG. 4, the flow path structure **11** is provided with a coupling pipe **11a**, a coupling pipe **11b**, a coupling pipe **11c**, a coupling pipe **11d**, and a hole **11e**.

(56) Here, although not shown in the drawings, flow paths such as a first supply flow path CC1, a second supply flow path CC2, a first discharge flow path CM1, and a second discharge flow path CM2 are provided in the flow path structure **11**. The first supply flow path CC1 is a flow path for supply of the first ink to the plurality of head main bodies **14**, the first ink being introduced into the coupling pipe **11a**. The second supply flow path CC2 is a flow path for supply of the second ink to the plurality of head main bodies **14**, the second ink being introduced into the coupling pipe **11b**. A filter used to capture foreign substances or the like is installed in an intermediate portion of each of the supply flow paths. The first discharge flow path CM1 is a flow path for discharge of the first ink from the plurality of head main bodies **14**. The second discharge flow path CM2 is a flow path for discharge of the second ink from the plurality of head main bodies **14**.

(57) The coupling pipes **11a**, **11b**, **11c**, and **11d** are pipe bodies protruding in the Z1 direction. More specifically, the coupling pipe **11a** is a pipe body that constitutes a flow path for supply of the first ink to the first supply flow path CC1. In addition, the coupling pipe **11b** is a pipe body that constitutes a flow path for supply of the second ink to the second supply flow path CC2. Meanwhile, the coupling pipe **11c** is a pipe body that constitutes a flow path for discharge of the first ink from the first discharge flow path CM1. In addition, the coupling pipe **11d** is a pipe body that constitutes a flow path for discharge of the second ink from the second discharge flow path CM2. The hole **11e** is a hole for insertion of a connector **12c** which will be described later.

(58) The wiring substrate **12** is a mounted component for electrical coupling between the plurality of head main bodies **14** and a collective substrate **16b** which will be described later. The wiring substrate **12** is, for example, a rigid wiring substrate. The wiring substrate **12** is disposed between the flow path structure **11** and the holder **13**, and the connector **12c** is installed on a surface of the wiring substrate **12** that faces the flow path structure **11**. The connector **12c** is a coupling component coupled to the collective substrate **16b** which will be described later. In addition, the wiring substrate **12** is provided with a plurality of holes **12a** and a plurality of opening portions **12b**. Each hole **12a** is a hole for allowance of coupling between the flow path structure **11** and the holder **13**. Each opening portion **12b** is a hole through which a wiring substrate **14h** that couples the head main body **14** and the wiring substrate **12** passes. The wiring substrate **14h** is coupled to a surface of the wiring substrate **12** that faces the Z1 direction. The wiring substrate **14h** is a member including a wire electrically coupled to a piezoelectric element PZ which will be described later, and is, for example, an FPC, a COF, an FFC, or the like. The “FPC” is an abbreviation for “Flexible Printed Circuits”. The “COF” is an abbreviation for “Chip On Film”. The “FFC” is an abbreviation for “Flexible Flat Cable”.

(59) The holder **13** is a structure that accommodates and supports the plurality of head main bodies

14. The holder **13** is formed of, for example, a resin material, a metal material, or the like. The holder **13** has a plate-like shape that extends in directions perpendicular to the Z-axis. In addition, the holder **13** is provided with a coupling pipe **13a**, a coupling pipe **13b**, a plurality of coupling pipes **13c**, a plurality of coupling pipes **13d**, and a plurality of wiring holes **13e**. In addition, although not shown in the drawing, a surface of the holder **13** that faces the Z2 direction is provided with a plurality of recess portions for accommodation of the plurality of head main bodies **14**.

(60) In the first embodiment, the holder **13** holds the six head main bodies **14_1** to **14_6**. The head main bodies **14** are arranged in the X2 direction in the order of the head main bodies **14_1**, **14_2**, **14_3**, **14_4**, **14_5**, and **14_6**. Here, the head main bodies **14_1** to **14_3** are disposed at positions offset from the head main bodies **14_4** to **14_6** in the Y1 direction. However, the head main bodies **14_1** to **14_6** partially overlap with each other as seen in the X1 direction or the X2 direction. In addition, arrangement directions DN (which will be described later) of the plurality of nozzles N of the head main bodies **14_1** to **14_6** are parallel to each other. Furthermore, each of the head main bodies **14_1** to **14_6** is disposed such that the arrangement direction DN is inclined with respect to the direction DM which is a transport direction of the medium PP.

(61) Here, although not shown in FIG. 4, a plurality of flow paths, which will be described later with reference to FIGS. 8 and 9, are provided inside the holder **13**. The plurality of flow paths include a first distribution supply flow path SP1, a second distribution supply flow path SP2, a plurality of first individual discharge flow paths DS1, a plurality of second individual discharge flow paths DS2, and a plurality of bypass flow paths BP. The first distribution supply flow path SP1 is a flow path that includes branches for supply of the first ink to the plurality of head main bodies **14**. The second distribution supply flow path SP2 is a flow path that includes branches for supply of the second ink to the plurality of head main bodies **14**. The first individual discharge flow path DS1 is provided for each of the head main bodies **14** that discharge the first ink, and is a flow path for introduction of the first ink to the first discharge flow path CM1 of the flow path structure **11**, the first ink being discharged from the head main bodies **14**. The second individual discharge flow path DS2 is provided for each of the head main bodies **14** that discharge the second ink, and is a flow path for introduction of the second ink to the second discharge flow path CM2 of the flow path structure **11**, the second ink being discharged from the head main bodies **14**. Two bypass flow paths BP are provided for each of the head main bodies **14** and the bypass flow paths BP are bypass flow paths through which a first common liquid chamber R1 and a second common liquid chamber R2 (which will be described later) communicate with each other. Note that, the flow paths of the holder **13** will be described based on FIGS. 8 to 10 which will be used for description later.

(62) In the first embodiment, regarding the head main bodies **14_1** to **14_6**, the first ink is supplied to the head main bodies **14_1** to **14_3** and the second ink is supplied to the head main bodies **14_4** to **14_6**.

(63) The coupling pipes **13a**, **13b**, **13c**, and **13d** are pipe-shaped protrusions protruding in the Z1 direction. More specifically, the coupling pipe **13a** is a pipe body that constitutes a flow path for supply of the first ink to the first distribution supply flow path SP1, and the coupling pipe **13a** communicates with the first supply flow path CC1 of the flow path structure **11**. In addition, the coupling pipe **13b** is a pipe body that constitutes a flow path for supply of the second ink to the second distribution supply flow path SP2, and the coupling pipe **13b** communicates with the second supply flow path CC2 of the flow path structure **11**. Meanwhile, the coupling pipes **13c** are pipe bodies that constitute flow paths for discharge of the first ink from the first individual discharge flow paths DS1 and the coupling pipes **13c** communicate with the first discharge flow path CM1 of the flow path structure **11**. In addition, the coupling pipes **13d** are pipe bodies that constitute flow paths for discharge of the second ink from the second individual discharge flow paths DS2 and the coupling pipes **13d** communicate with the second discharge flow path CM2 of the flow path structure **11**. The wiring holes **13e** are holes through which the wiring substrates **14h** that couple

the head main bodies **14** and the wiring substrate **12** pass.

(64) Each head main body **14** ejects ink. Specifically, although not shown in FIG. **4**, each head main body **14** includes M nozzles N for ejection of the first ink or the second ink. The nozzles N are provided in the nozzle surface FN, which is a surface of each head main body **14** that faces the Z2 direction. Details of the head main body **14** will be described based on FIG. **6** which will be used for description later.

(65) The fixation plate **15** is a plate member for fixation of the plurality of head main bodies **14** to the holder **13**. Specifically, the fixation plate **15** is disposed in a state where the plurality of head main bodies **14** are interposed between the fixation plate **15** and the holder **13**, and is fixed to the holder **13** by means of an adhesive. The fixation plate **15** is formed of, for example, a metal material. The fixation plate **15** is provided with a plurality of opening portions **15a** for exposure of the nozzles N of the plurality of head main bodies **14**. In an example shown in FIG. **4**, the plurality of opening portions **15a** are provided such that the opening portion **15a** is provided for each of the head main bodies **14**. Note that, the opening portion **15a** may be shared by two or more head main bodies **14**.

(66) The base **16** is a member for fixation of the flow path structure **11**, the wiring substrate **12**, the holder **13**, the plurality of head main bodies **14**, and the fixation plate **15** to the above-described supporting body **41**. The base **16** includes a main body **16a**, a collective substrate **16b**, and a cover **16c**.

(67) The main body **16a** is fixed to the holder **13** by being screwed or the like so that the flow path structure **11** and the wiring substrate **12** disposed between the base **16** and the holder **13** are held by the main body **16a**. The main body **16a** is formed of, for example, a resin material. The main body **16a** includes a plate-shaped portion that faces a plate-shaped portion of the above-described flow path structure **11** and the plate-shaped portion is provided with a plurality of holes **16d** into which the above-described coupling pipes **11a**, **11b**, **11c**, and **11d** are inserted. In addition, the main body **16a** includes a portion that extends in the Z2 direction from the plate-shaped portion, and a tip end of that portion is provided with a flange **16e** for fixation to the above-described supporting body **41**.

(68) The collective substrate **16b** is a mounted component for electrical coupling between the control module **120** and the above-described wiring substrate **12**. The collective substrate **16b** is, for example, a rigid wiring substrate. The cover **16c** is a plate-shaped member for protection of the collective substrate **16b** and for fixation of the collective substrate **16b** to the main body **16a**. The cover **16c** is formed of, for example, a resin material or the like, and the cover **16c** is fixed to the main body **16a** by being screwed or the like.

(69) 1-4. Head Main Body **14**

(70) FIG. **5** is a plan view schematically showing flow paths of the head main body **14**. For the sake of convenience, the following description will be made while appropriately using a V-axis and a W-axis in addition to the X-axis, the Y-axis, and the Z-axis. In addition, a direction extending along the V-axis is a V1 direction, and a direction opposite to the V1 direction is a V2 direction. Similarly, directions that extend along the W-axis and are opposite to each other are a W1 direction and a W2 direction.

(71) Here, the V-axis is an axis along the arrangement direction DN of the plurality of nozzles N, which will be described later, and is an axis obtained by rotating the Y-axis around the Z-axis by a predetermined angle. The W-axis is an axis obtained by rotating the X-axis around the Z-axis by the predetermined angle. Therefore, the V-axis and the W-axis are typically orthogonal to each other. However, the present disclosure is not limited thereto. For example, the V-axis and the W-axis only need to intersect each other at an angle within a range of 80 degrees or more and 100 degrees or less. In addition, the predetermined angle (that is, the angle formed between the V-axis and the Y-axis, or the angle formed between the W-axis and the X-axis) is, for example, in a range of 40 degrees or more and 60 degrees or less.

(72) As shown in FIG. 5, the head main body **14** is provided with M nozzles N, M individual flow paths PJ, the first common liquid chamber R1, and the second common liquid chamber R2. Here, the first common liquid chamber R1 and the second common liquid chamber R2 communicate with each other via the M individual flow paths PJ. In addition, as represented by two-dot chain lines in FIG. 5, the bypass flow paths BP1 and BP2 are coupled to the first common liquid chamber R1 and the second common liquid chamber R2. Hereinafter, the bypass flow paths BP1 and BP2 may be collectively referred to as the bypass flow paths BP. The bypass flow paths BP1 and BP2 are flow paths that bypass the M individual flow paths PJ and through which the first common liquid chamber R1 and the second common liquid chamber R2 communicate with each other, the bypass flow paths BP1 and BP2 being provided in the holder **13**. Ink supplied to the first common liquid chamber R1 flows into any one of the M individual flow paths PJ and the bypass flow paths BP1 and BP2. Ink not ejected from the respective nozzles N of the M individual flow paths PJ and ink flowing through the bypass flow paths BP1 and BP2 are discharged to the second common liquid chamber R2. The flow path resistance of the bypass flow paths BP is smaller than the flow path resistance of each of the M individual flow paths PJ. Since the liquid ejection head **10** includes the bypass flow paths BP, the liquid ejection head **10** can retrieve air bubbles in ink. Details of the bypass flow paths BP1 and BP2 will be described based on FIGS. 8, 9, and 10 which will be used for description later.

(73) The head main body **14** includes a surface that faces the medium PP, and as shown in FIG. 5, the surface is provided with the M nozzles N. The plurality of nozzles N are arranged along the V-axis. Each of the M nozzles N ejects ink in the Z2 direction.

(74) Here, a group of the plurality of nozzles N constitutes a nozzle row Ln. Further, the plurality of nozzles N are arranged at substantially equal intervals at a predetermined pitch. The predetermined pitch is a distance between the centers of the plurality of nozzles N in a direction along the V-axis.

(75) The individual flow path PJ communicates with each of the M nozzles N. The M individual flow paths PJ extend along the W-axis and communicate with different nozzles N. The M individual flow paths PJ are arranged along the V-axis.

(76) As shown in FIG. 5, each of the M individual flow paths PJ includes a pressure chamber Ca, a pressure chamber Cb, a nozzle flow path Nf, an individual supply flow path Ra1, an individual discharge flow path Ra2, a first communication flow path Na1, a second communication flow path Na2, a narrowed portion Ap1, and a narrowed portion Ap2.

(77) Each of the pressure chamber Ca and the pressure chamber Cb in any of the M individual flow paths PJ is a space that extends along the W-axis and in which ink to be ejected from the nozzle N communicating with the individual flow path PJ is stored. In an example shown in FIG. 5, M pressure chambers Ca are arranged along the V-axis. Similarly, M pressure chambers Cb are arranged along the V-axis. Note that, in each of the M individual flow paths PJ, the positions of the pressure chamber Ca and the pressure chamber Cb in the direction along the V-axis are the same as each other in the example shown in FIG. 5. However, the positions thereof in the direction along the V-axis may be different from each other. Note that, hereinafter, when the pressure chamber Ca and the pressure chamber Cb are not to be particularly distinguished, each of the pressure chamber Ca and the pressure chamber Cb may be collectively referred to as a “pressure chamber C”.

(78) The nozzle flow path Nf is disposed between the pressure chamber Ca and the pressure chamber Cb in each of the M individual flow paths PJ. Here, the pressure chamber Ca communicates with the nozzle flow path Nf via the first communication flow path Na1 extending along the Z-axis. The pressure chamber Cb communicates with the nozzle flow path Nf via the second communication flow path Na2 extending along the Z-axis.

(79) In the M individual flow paths PJ, the nozzle flow paths Nf are spaces extending along the W-axis. In addition, M nozzle flow paths Nf are arranged at intervals along the V-axis. Each of the M nozzle flow paths Nf is provided with the nozzle N. Ink is ejected from the nozzle N in each of the

M nozzle flow paths Nf when there is a change in pressures in the pressure chamber Ca and the pressure chamber Cb.

(80) Each of the first communication flow path Na1 and the second communication flow path Na2 is a space extending along the Z-axis.

(81) Here, the first common liquid chamber R1 and the second common liquid chamber R2 communicate with the M individual flow paths PJ. The pressure chamber Ca communicates with the first common liquid chamber R1 via the narrowed portion Ap1 extending along the W-axis and the individual supply flow path Ra1 extending along the Z-axis. The pressure chamber Cb communicates with the second common liquid chamber R2 via the narrowed portion Ap2 extending along the W-axis and the individual discharge flow path Ra2 extending along the Z-axis.

(82) The narrowed portion Ap1 is a flow path provided between the pressure chamber Ca and the individual supply flow path Ra1. The narrowed portion Ap2 is a flow path provided between the pressure chamber Cb and the individual discharge flow path Ra2. Hereinafter, each of the narrowed portion Ap1 and the narrowed portion Ap2 may be collectively referred to as a “narrowed portion Ap”. The narrowed portions Ap are flow paths formed to be narrower than other regions in the individual flow path PJ. The other regions in the individual flow path PJ are the pressure chamber Ca and the pressure chamber Cb. More specifically, the cross-sectional area of the narrowed portion Ap1 is smaller than the cross-sectional area of the pressure chamber Ca. Therefore, the narrowed portion Ap1 has a higher flow path resistance than the pressure chamber Ca. Similarly, the cross-sectional area of the narrowed portion Ap2 is smaller than the cross-sectional area of the pressure chamber Cb. Therefore, the narrowed portion Ap2 has a higher flow path resistance than the pressure chamber Cb. Furthermore, the narrowed portions Ap may be formed to be narrower than all other regions in the individual flow path PJ. The all other regions in the individual flow path PJ are the pressure chamber Ca, the pressure chamber Cb, the nozzle flow path Nf, the individual supply flow path Ra1, the individual discharge flow path Ra2, the first communication flow path Na1, and the second communication flow path Na2. Since the narrowed portions Ap are formed to be narrower than the other regions in the individual flow path PJ, the flow path resistances of the narrowed portions Ap are set to be higher than that of the other regions in the individual flow path PJ.

(83) Each of the first common liquid chamber R1 and the second common liquid chamber R2 is a space that extends along the V-axis to overlap with the M individual flow paths PJ as seen along the W-axis. The first common liquid chamber R1 is coupled to an end, in the W2 direction, of each of the M individual flow paths PJ. In the first common liquid chamber R1, ink to be supplied to the plurality of individual flow paths PJ is stored. Meanwhile, the second common liquid chamber R2 is coupled to ends, in the W1 direction, of the individual flow paths PJ. In the second common liquid chamber R2, ink that is discharged from the plurality of individual flow paths PJ without being ejected from the nozzles N is stored.

(84) The first common liquid chamber R1 is provided with a supply port IO1, a discharge port IO3a, and a discharge port IO3b. The supply port IO1 is a pipeline for introduction of ink into the first common liquid chamber R1 from a distribution supply flow path SP of the holder 13. The discharge port IO3a is a pipeline for discharge of ink from the first common liquid chamber R1 to the bypass flow path BP1. The discharge port IO3b is a pipeline for discharge of ink from the first common liquid chamber R1 to the bypass flow path BP2. Note that, the distribution supply flow path SP is a general term for the first distribution supply flow path SP1 or the second distribution supply flow path SP2, which will be described later.

(85) Here, the distribution supply flow path SP is coupled to the circulation mechanism 150 via the first supply flow path CC1 or the second supply flow path CC2 of the flow path structure 11. Therefore, a flow path from the coupling pipe 11a or the coupling pipe 11b to the first common liquid chamber R1 is provided in common for the M pressure chambers Ca and the flow path constitutes a common supply flow path CF1 for supply of ink to the M individual flow paths PJ.

(86) The second common liquid chamber R2 is provided with a discharge port IO2, an introduction port IO4a, and an introduction port IO4b. The discharge port IO2 is a pipeline for discharge of ink from the second common liquid chamber R2 to an individual discharge flow path DS of the holder 13. The introduction port IO4a is a pipeline for introduction of ink from the bypass flow path BP1 into the second common liquid chamber R2. The introduction port IO4b is a pipeline for introduction of ink from the bypass flow path BP2 into the second common liquid chamber R2. Note that, the individual discharge flow path DS is the first individual discharge flow path DS1 or the second individual discharge flow path DS2, which will be described later.

(87) Here, the individual discharge flow path DS is coupled to the circulation mechanism 150 via the first discharge flow path CM1 or the second discharge flow path CM2 of the flow path structure 11. Therefore, a flow path from the second common liquid chamber R2 to the coupling pipe 11a or the coupling pipe 11b is provided in common for the M pressure chambers Cb and the flow path constitutes a common discharge flow path CF2 for discharge of ink from the M individual flow paths PJ.

(88) FIG. 6 is a cross-sectional view of the head main body 14. FIG. 6 shows a cross section of the head main body 14 cut by a plane including the W-axis and the Z-axis. As shown in FIG. 6, the head main body 14 includes a nozzle substrate 14a, a flow path substrate 14b, a pressure chamber substrate 14c, a vibration plate 14d, 2×M piezoelectric elements PZ, a case 14f, a protection plate 14g, the wiring substrate 14h, and a vibration absorbing body 14j.

(89) The nozzle substrate 14a, the flow path substrate 14b, the pressure chamber substrate 14c, and the vibration plate 14d are stacked in this order in the Z1 direction. Each of these members extends along the V-axis and is manufactured, for example, by processing a silicon single crystal substrate with a semiconductor processing technology. In addition, these members are bonded to each other by means of an adhesive or the like. Note that, another layer such as an adhesive layer or a substrate may be appropriately interposed between two of these members that are adjacent to each other.

(90) The nozzle substrate 14a is provided with the M nozzles N. Each of the M nozzles N is a through-hole that penetrates the nozzle substrate 14a and through which ink passes. The M nozzles N are arranged in the direction along the V-axis.

(91) The flow path substrate 14b is provided with a portion of each of the first common liquid chamber R1 and the second common liquid chamber R2 and is provided with portions of the M individual flow paths PJ excluding the pressure chambers Ca, the narrowed portions Ap1, the pressure chambers Cb, and the narrowed portions Ap2. That is, the flow path substrate 14b is provided with the nozzle flow paths Nf, the first communication flow paths Na1, the second communication flow paths Na2, the individual supply flow paths Ra1, and the individual discharge flow paths Ra2.

(92) The portion of each of the first common liquid chamber R1 and the second common liquid chamber R2 is a space penetrating the flow path substrate 14b. The vibration absorbing body 14j that closes an opening attributable to the space is installed on a surface of the flow path substrate 14b that faces the Z2 direction.

(93) The vibration absorbing body 14j is a layer-shaped member formed of an elastic material. The vibration absorbing body 14j constitutes a portion of a wall surface of each of the first common liquid chamber R1 and the second common liquid chamber R2, and absorbs a pressure change in the first common liquid chamber R1 and the second common liquid chamber R2.

(94) The nozzle flow paths Nf are spaces in grooves provided at a surface of the flow path substrate 14b that faces the Z2 direction. Here, the nozzle substrate 14a constitutes portions of wall surfaces of the nozzle flow paths Nf.

(95) Each of the first communication flow paths Na1 and the second communication flow paths Na2 is a space penetrating the flow path substrate 14b.

(96) Each of the individual supply flow paths Ra1 and the individual discharge flow paths Ra2 is a

space penetrating the flow path substrate **14b**. The first common liquid chamber **R1** and the narrowed portions **Ap1** communicate with each other through the individual supply flow paths **Ra1** and ink from the first common liquid chamber **R1** is supplied to the pressure chambers **Ca** via the narrowed portions **Ap1**. Here, one end of each individual supply flow path **Ra1** is open at a surface of the flow path substrate **14b** that faces the **Z1** direction. Meanwhile, the other end of each individual supply flow path **Ra1** is an upstream end of the individual flow path **PJ** and is open at the wall surface of the first common liquid chamber **R1** in the flow path substrate **14b**. With regard to this, the second common liquid chamber **R2** and the narrowed portions **Ap2** communicate with each other through the individual discharge flow paths **Ra2** and ink from the pressure chambers **Cb** that is discharged via the narrowed portions **Ap2** is discharged to the second common liquid chamber **R2** through the individual discharge flow paths **Ra2**. Here, one end of each individual discharge flow path **Ra2** is open at a surface of the flow path substrate **14b** that faces the **Z1** direction. Meanwhile, the other end of each individual discharge flow path **Ra2** is a downstream end of the individual flow path **PJ** and is open at the wall surface of the second common liquid chamber **R2** in the flow path substrate **14b**.

(97) The pressure chamber substrate **14c** is provided with the pressure chamber **Ca**, the narrowed portion **Ap1**, the pressure chamber **Cb**, and the narrowed portion **Ap2** of each of the **M** individual flow paths **PJ**. Each of the pressure chamber **Ca**, the narrowed portion **Ap1**, the pressure chamber **Cb**, and the narrowed portion **Ap2** penetrates the pressure chamber substrate **14c** and is a gap between the flow path substrate **14b** and the vibration plate **14d**.

(98) The vibration plate **14d** is a plate-shaped member that can elastically vibrate. The vibration plate **14d** is, for example, a laminated body including a first layer formed of silicon oxide and a second layer formed of zirconium oxide. Here, another layer such as a metal oxide layer may be interposed between the first layer and the second layer. Note that, a portion of the vibration plate **14d** or the entire vibration plate **14d** may be integrally formed with the pressure chamber substrate **14c** while being formed of the same material as the pressure chamber substrate **14c**. For example, the vibration plate **14d** and the pressure chamber substrate **14c** can be integrally formed with each other by selectively removing portions, in a thickness direction, of regions of a plate-shaped member having a predetermined thickness, the regions corresponding to the pressure chambers **C**. In addition, the vibration plate **14d** may be composed of a single-material layer.

(99) On a surface of the vibration plate **14d** that faces the **Z1** direction, **M** piezoelectric elements **PZa** respectively corresponding to the **M** pressure chambers **Ca** and **M** piezoelectric elements **PZb** respectively corresponding to the **M** pressure chambers **Cb** are installed. In the following description, “piezoelectric element **PZ**” is a general term for the piezoelectric elements **PZa** and the piezoelectric elements **PZb**. In addition, the piezoelectric element **PZ** corresponding to the pressure chamber **C** means the piezoelectric element **PZ** that overlaps a portion of the pressure chamber **C** or the entire pressure chamber **C** in plan view along the **Z**-axis.

(100) FIG. 7 is an enlarged view of the vicinities of the piezoelectric elements **PZ** of FIG. 6. As shown in FIG. 7, the **M** piezoelectric elements **PZa** are formed by stacking a common electrode **Qua** and individual electrodes **Qda** that face each other and piezoelectric bodies **Qma** that are disposed between the common electrode **Qua** and the individual electrodes **Qda**. The common electrode **Qua** is provided in common for the **M** pressure chambers **Ca**. The individual electrodes **Qda** are respectively provided for the **M** pressure chambers **Ca**. Similarly, the **M** piezoelectric elements **PZb** are formed by stacking a common electrode **Qub** and individual electrodes **Qdb** that face each other and piezoelectric bodies **Qmb** that are disposed between the common electrode **Qub** and the individual electrodes **Qdb**. The common electrode **Qub** is provided in common for the **M** pressure chambers **Cb**. The individual electrodes **Qdb** are respectively provided for the **M** pressure chambers **Cb**.

(101) As shown in FIG. 7, the common electrode **Qua** is provided on surfaces of the piezoelectric bodies **Qma** that face the **Z1** direction. The common electrode **Qub** is provided on surfaces of the

piezoelectric bodies Qmb that face the Z1 direction. In the following description, each of the common electrode Qua and the common electrode Qub may be collectively referred to as a common electrode Qu. The individual electrodes Qda are provided on surfaces of the piezoelectric bodies Qma that face the Z2 direction. The individual electrodes Qdb are provided on surfaces of the piezoelectric bodies Qmb that face the Z2 direction. In the following description, each of the individual electrodes Qda and the individual electrodes Qdb may be collectively referred to as an individual electrode Qd. A predetermined reference potential Vbs is supplied to the common electrodes Qu, and the drive signal Com is supplied to the individual electrodes Qd. In the first embodiment, the common electrodes Qu are so-called upper electrodes, and the individual electrodes Qd are so-called lower electrodes. A configuration in which the common electrodes Qu are lower electrodes and the individual electrodes Qd are upper electrodes may also be adopted. (102) Each of the $2 \times M$ piezoelectric elements PZ causes a change in pressure of ink in the pressure chamber C corresponding thereto so that the ink in the pressure chamber C is ejected from the nozzle N. The piezoelectric elements PZ causes the vibration plate 14d to vibrate in accordance with deformation thereof when the drive signal Com is supplied thereto. Because of the vibration, the pressure chambers C expand and contract and thus the pressure of ink in the pressure chambers C changes.

(103) The description will be made referring again to FIG. 6. The case 14f is a case for storage of ink. The case 14f is provided with a space constituting a remaining portion of each of the first common liquid chamber R1 and the second common liquid chamber R2 other than a portion provided in the flow path substrate 14b.

(104) The protection plate 14g is a plate-shaped member installed on the surface of the vibration plate 14d that faces the Z1 direction, protects the $2 \times M$ piezoelectric elements PZ, and reinforces the mechanical strength of the vibration plate 14d. Here, a space for accommodation of the $2 \times M$ piezoelectric elements PZ is formed between the protection plate 14g and the vibration plate 14d.

(105) The wiring substrate 14h is mounted on the surface of the vibration plate 14d that faces the Z1 direction, and is a mounted component for electrical coupling between the control module 120 and the head main body 14. A drive circuit 14i is mounted on the wiring substrate 14h. The drive circuit 14i includes the switching circuit 141 shown in FIG. 1.

(106) In the head main body 14 configured as described above, because of the operation of the circulation mechanism 150, ink flows through the first common liquid chamber R1, the individual supply flow paths Ra1, the narrowed portions Ap1, the pressure chambers Ca, the first communication flow paths Na1, the nozzle flow paths Nf, the second communication flow paths Na2, the pressure chambers Cb, the narrowed portions Ap2, the individual discharge flow paths Ra2, and the second common liquid chamber R2 in this order.

(107) As shown in FIG. 6, one ejector D includes two piezoelectric elements PZ, two pressure chambers C, and one nozzle N. When the drive signal Com is supplied to the two piezoelectric elements PZ based on the designation signal SI, the ejector D drives the two piezoelectric elements PZ by means of the drive signal Com so that ink in the two pressure chambers C is ejected from the nozzle N. In the following description, two piezoelectric elements PZ included in the ejector D[m] may be described as “piezoelectric elements PZ[m]”, with respect to any m ranging from 1 to M. Further, the individual electrodes Qd respectively provided in the two piezoelectric elements PZ may be described as “individual electrodes Qd[m]”. The individual flow path PJ corresponding to the ejector D[m] (in other words, the individual flow path PJ including a pressure chamber Ca[m] and a pressure chamber Cb[m]) may be described as an “individual flow path PJ[m]”. In addition, each element in the individual flow path PJ may be described with a suffix “[m]” added thereto.

(108) An individual flow path PJ[m1] of a certain head main body 14 is an example of a “first individual flow path”, with respect to any integer m1 ranging from 1 to M. A pressure chamber Ca[m1] included in the individual flow path PJ[m1] is an example of a “first pressure chamber”, and a piezoelectric element PZa[m1] is an example of a “first piezoelectric element”. A pressure

chamber Cb[m1] included in the individual flow path PJ[m1] is an example of a “second pressure chamber”, and a piezoelectric element PZb[m1] is an example of a “second piezoelectric element”. A nozzle N[m1] communicating with the individual flow path PJ[m1] is an example of a “first nozzle”. A narrowed portion Ap1[m1] included in the individual flow path PJ[m1] is an example of a “first narrowed portion”. A narrowed portion Ap2[m1] included in the individual flow path PJ[m1] is an example of a “second narrowed portion”.

(109) An individual flow path PJ[m2] is an example of a “second individual flow path”, with respect to an integer m2 different from the integer m1 in a range from 1 to M. A pressure chamber Ca[m2] included in the individual flow path PJ[m2] is an example of a “third pressure chamber”, and a piezoelectric element PZa[m2] is an example of a “third piezoelectric element”. A pressure chamber Cb[m2] included in the individual flow path PJ[m2] is an example of a “fourth pressure chamber”, and a piezoelectric element PZb[m2] is an example of a “fourth piezoelectric element”. A nozzle N[m2] communicating with the individual flow path PJ[m2] is an example of a “second nozzle”. A narrowed portion Ap1[m2] included in the individual flow path PJ[m2] is an example of a “third narrowed portion”. A narrowed portion Ap2[m2] included in the individual flow path PJ[m2] is an example of a “fourth narrowed portion”.

(110) 1-5. Holder **13**

(111) FIG. **8** is a plan view of the holder **13**. FIG. **9** is a perspective view showing the flow paths provided in the holder **13** and the head main bodies **14**. Note that, in FIG. **8**, an example of the internal structure of the holder **13** as seen in the Z2 direction is shown by broken lines. In FIG. **9**, the fixation plate **15** is shown in addition to the flow paths of the holder **13** and the plurality of head main bodies **14**.

(112) As shown in FIGS. **8** and **9**, in the holder **13**, the first distribution supply flow path SP1, the second distribution supply flow path SP2, three first individual discharge flow paths DS1, three second individual discharge flow paths DS2, six bypass flow paths BP1, and six bypass flow paths BP2 are provided.

(113) The first distribution supply flow path SP1 is a flow path including three branch portions for supply of the first ink to three head main bodies **14**, the first ink being introduced into the coupling pipe **13a**. The second distribution supply flow path SP2 is a flow path including three branch portions for supply of the second ink to three head main bodies **14**, the second ink being introduced into the coupling pipe **13b**.

(114) The first individual discharge flow path DS1 is provided for each of the head main bodies **14** that use the first ink, and is a flow path for discharge of the first ink from the coupling pipe **13c**, the first ink being introduced from the corresponding head main body **14**. The second individual discharge flow path DS2 is provided for each of the head main bodies **14** that use the second ink, and is a flow path for discharge of the second ink from the coupling pipes **13d**, the second ink being introduced from the corresponding head main body **14**.

(115) The bypass flow path BP1 and the bypass flow path BP2 are provided for each of the head main bodies **14** and the bypass flow path BP1 and the bypass flow path BP2 are flow paths through which the first common liquid chamber R1 and the second common liquid chamber R2 described above communicate with each other. However, the bypass flow path BP1 and the bypass flow path BP2 are positioned on opposite sides to each other with respect to the center of the first common liquid chamber R1 or the second common liquid chamber R2 in the direction along the X-axis. In an example shown in FIG. **8**, the bypass flow path BP1 is positioned closer to a side to which the V2 direction extends than the bypass flow path BP2 is. In addition, each of the bypass flow path BP1 and the bypass flow path BP2 has a U-like shape as seen in a direction along the Z-axis.

(116) FIG. **10** is a sectional view taken along line X-X in FIG. **8**. In FIG. **10**, the head main body **14** and the fixation plate **15** are shown in addition to the holder **13**. As shown in FIG. **10**, the holder **13** has a plate-like shape that extends in directions perpendicular to the Z-axis. The holder **13** includes a layer **31** and a layer **32** and the layer **31** and the layer **32** are stacked in this order in the

Z2 direction. Each of the layer **31** and the layer **32** is formed of, for example, a resin material and is formed through injection molding. The layer **31** and the layer **32** are bonded to each other by means of, for example, an adhesive.

(117) Each of the above-described flow paths that the holder **13** includes is provided in a laminated body composed of the layer **31** and the layer **32**, and a recess portion **13f** for accommodation of the head main body **14** is provided at a surface of the layer **32** that faces the **Z2** direction. In an example shown in FIG. **10**, the thickness of the layer **32** is larger than the thickness of the layer **31**. Therefore, the thickness of the layer **32** required for formation of the recess portion **13f** can be easily secured.

(118) Here, the first distribution supply flow path **SP1** includes a vertical flow path **SPa** and a horizontal flow path **SPb**. The vertical flow path **SPa** extends in the direction along the **Z**-axis and is composed of a hole that penetrates the layer **32**. The horizontal flow path **SPb** extends in a direction orthogonal to the **Z**-axis, and is provided between the layer **31** and the layer **32**. In the example shown in FIG. **10**, the horizontal flow path **SPb** is composed of a groove provided at a surface of the layer **31** that faces the **Z2** direction and a groove provided at a surface of the layer **32** that faces the **Z1** direction. Note that, although not shown in FIG. **10**, the second distribution supply flow path **SP2** is configured in the same manner as the first distribution supply flow path **SP1**.

(119) The bypass flow path **BP1** includes a first portion **BP1a**, a second portion **BP1b**, and a third portion **BP1c**. Each of the first portion **BP1a** and the second portion **BP1b** extends in the direction along the **Z**-axis and is composed of a hole that penetrates the layer **32**. The third portion **BP1c** extends in a direction orthogonal to the **Z**-axis, and is provided between the layer **31** and the layer **32**. In the example shown in FIG. **10**, the third portion **BP1c** is composed of a groove provided at the surface of the layer **31** that faces the **Z2** direction and a groove provided at the surface of the layer **32** that faces the **Z1** direction.

(120) Similarly, the bypass flow path **BP2** includes a first portion **BP2a**, a second portion **BP2b**, and a third portion **BP2c**. Each of the first portion **BP2a** and the second portion **BP2b** extends in the direction along the **Z**-axis and is composed of a hole that penetrates the layer **32**. The third portion **BP2c** extends in a direction orthogonal to the **Z**-axis, and is provided between the layer **31** and the layer **32**. In the example shown in FIG. **10**, the third portion **BP2c** is composed of a groove provided at the surface of the layer **31** that faces the **Z2** direction and a groove provided at the surface of the layer **32** that faces the **Z1** direction.

(121) The two bypass flow paths **BP** are formed to be wider than each of the **M** individual flow paths **PJ**. Therefore, the two bypass flow paths **BP** have lower flow path resistances than each of the **M** individual flow paths **PJ**. In addition, the two bypass flow paths **BP** are formed to be narrower than each of the first common liquid chamber **R1** and the second common liquid chamber **R2**. In addition, as described above, the two bypass flow paths **BP** are formed to be bent. Therefore, the flow path resistance of the two bypass flow paths **BP** is greater than those of the first common liquid chamber **R1** and the second common liquid chamber **R2**.

(122) 1-6. Configuration of Head Main Body **14**

(123) FIG. **11** is a block diagram showing an example of the configuration of the head main body **14**. The liquid ejection head **10** include, in addition to the switching circuit **141** described above, an internal wire **LHa** to which the drive signal **Com-A** from the drive signal generation circuit **190** is supplied, an internal wire **LHb** to which the drive signal **Com-B** from the drive signal generation circuit **190** is supplied, and an internal wire **LHd** to which the reference potential **Vbs** is supplied.

(124) As shown in FIG. **11**, the switching circuit **141** includes **M** switches **SW1a[1]** to **SW1a[M]**, **M** switches **SW2a[1]** to **SW2a[M]**, **M** switches **SW1b[1]** to **SW1b[M]**, **M** switches **SW2b[1]** to **SW2b[M]**, and a coupling state designation circuit **142** that designates the state of coupling of each switch. Note that, as each switch, for example, a transmission gate can be adopted.

(125) The coupling state designation circuit **142** generates, based on the designation signal **SI**

supplied from the control module 120 and at least a portion of a latch signal LAT, coupling state designation signals SL1a[1] to SL1a[M], coupling state designation signals SL2a[1] to SL2a[M], coupling state designation signals SL1b[1] to SL1b[M], and coupling state designation signals SL2b[1] to SL2b[M].

(126) A coupling state designation signal SL1a[m] is for designation of whether a switch SW1a[m] is to be turned on or off, with respect to any m ranging from 1 to M. A coupling state designation signal SL2a[m] is for designation of whether a switch SW2a[m] is to be turned on or off, with respect to any m ranging from 1 to M. A coupling state designation signal SL1b[m] is for designation of whether a switch SW1b[m] is to be turned on or off, with respect to any m ranging from 1 to M. A coupling state designation signal SL2b[m] is for designation of whether a switch SW2b[m] is to be turned on or off, with respect to any m ranging from 1 to M.

(127) The switch SW1a[m] switches, in accordance with the coupling state designation signal SL1a[m], between conduction and non-conduction between the internal wire LHa and individual electrodes Qda [m] of a piezoelectric element PZa[m] provided in an ejector D[m], with respect to any m ranging from 1 to M. For example, the switch SW1a[m] is turned on when the coupling state designation signal SL1a[m] is at a high level and is turned off when the coupling state designation signal SL1a[m] is at a low level.

(128) The switch SW2a[m] switches, in accordance with the coupling state designation signal SL2a[m], between conduction and non-conduction between the internal wire LHb and the individual electrodes Qda [m] of the piezoelectric element PZa[m] provided in the ejector D[m], with respect to any m ranging from 1 to M. For example, the switch SW2a[m] is turned on when the coupling state designation signal SL2a[m] is at a high level and is turned off when the coupling state designation signal SL2a[m] is at a low level.

(129) The switch SW1b[m] switches, in accordance with the coupling state designation signal SL1b[m], between conduction and non-conduction between the internal wire LHa and individual electrodes Qdb [m] of a piezoelectric element PZb[m] provided in the ejector D[m], with respect to any m ranging from 1 to M. For example, the switch SW1b[m] is turned on when the coupling state designation signal SL1b[m] is at a high level and is turned off when the coupling state designation signal SL1b[m] is at a low level.

(130) The switch SW2b[m] switches, in accordance with the coupling state designation signal SL2b[m], between conduction and non-conduction between the internal wire LHb and the individual electrodes Qdb [m] of the piezoelectric element PZb[m] provided in the ejector D[m], with respect to any m ranging from 1 to M. For example, the switch SW2b[m] is turned on when the coupling state designation signal SL2b[m] is at a high level and is turned off when the coupling state designation signal SL2b[m] is at a low level.

(131) In the first embodiment, it will be assumed that the waveforms of the drive signals Com supplied to two piezoelectric elements PZ included in one ejector D are approximately the same as each other. Here, “being approximately the same as each other” is a concept including a case where the waveforms can be considered as waveforms that are the same as each other when an error is taken into consideration, in addition to a case where the waveforms are completely the same as each other. For example, when both of the switch SW1a[m] and the switch SW1b[m] are on and both of the switch SW2a[m] and the switch SW2b[m] are off, the drive signal Com-A is supplied to two piezoelectric elements PZ included in the ejector D[m], with respect to any m ranging from 1 to M.

(132) 1-7. Operation in Comparative Embodiment

(133) As described above, ink in the first embodiment has a feature of being easily thickened. Therefore, ink thickened to a certain degree may stay inside the nozzles N or near the nozzles N before the printing ejection operation. Hereinafter, thickened ink will be referred to as “thickened ink”. Before the printing ejection operation, the nozzle surfaces FN are sealed by the cap 182 through the cap sealing operation, and basically, ink in the nozzles N and air are isolated from each

other. Therefore, ideally, thickening of the ink due to evaporation of a liquid component in the ink does not occur during the cap sealing operation. However, in practice, it is difficult to completely prevent contact between the ink in the nozzles N and the air even during the cap sealing operation. Therefore, in the case of the ink in the first embodiment, in a specific situation where the cap sealing operation is performed for a long time or an environmental temperature is high, thickened ink may stay inside the nozzles N or near the nozzles N even during the cap sealing operation.

(134) It may be conceivable that it is possible to eliminate a problem that the thickened ink stays by performing the circulation operation and the minute vibration operation. However, although depending on the order in which the circulation operation and the minute vibration operation are started, the thickened ink may continue to stay. Hereinafter, an embodiment in which the minute vibration operation is started first and then the circulation operation is started will be described as a “comparative embodiment”. The comparative embodiment will be described with reference to FIGS. 12 to 15.

(135) FIG. 12 is a view showing how the vicinity of the nozzle N is before the minute vibration operation is started in the comparative embodiment. FIG. 13 is a view showing how the vicinity of the nozzle N is after the minute vibration operation is started in the comparative embodiment. FIGS. 12 to 15 and FIGS. 17 to 20 (which will be used for description later) show enlarged views of the vicinity of the nozzle N in the cross-sectional view of the head main body 14 shown in FIG. 6. As shown in FIG. 12, thickened ink Bu stays inside and near the nozzle N. As shown in FIG. 13, when the minute vibration operation is started, a stream FRb along the Z-axis is generated in the nozzle N. As shown in FIG. 13, the minute vibration operation is performed to such a degree that the stream FRb reaches the nozzle flow path Nf. The stream FRb eliminates thickening at a position inside the nozzle N and at a position closer to a side to which the Z1 direction extends than the nozzle N is. However, as shown in FIG. 13, it is difficult for the stream FRb caused by the minute vibration operation to reach a position near the nozzle N in the nozzle flow path Nf in comparison with the position inside the nozzle N and at the position closer to the side to which the Z1 direction extends than the nozzle N is. Therefore, the thickened ink Bu may continue to stay at the position near the nozzle N.

(136) FIG. 14 is a view showing how the vicinity of the nozzle N is immediately after the circulation operation is started in the comparative embodiment. FIG. 15 is a view showing how the vicinity of the nozzle N is after the circulation operation is performed for a certain period in the comparative embodiment. As shown in FIG. 14, when the circulation operation is started, a stream FRc along the nozzle flow path Nf is generated.

(137) A pressure that is applied to ink in the individual flow path PJ per unit period because of the minute vibration operation is larger than a pressure that is applied to the ink in the individual flow path PJ per unit period because of the circulation operation. In order to make it easy to understand a difference between the magnitudes of the pressures, the stream FRb shown in FIGS. 14 and 15, and FIGS. 19 and 20 (which will be used for description later) is shown to be thicker than the stream FRc.

(138) As shown in FIG. 14, since the stream FRb reaches the nozzle flow path Nf, the stream FRc meanders because of the stream FRb at a position upstream of the vicinity of the nozzle N. Specifically, at the position upstream of the vicinity of the nozzle N, a direction in which the stream FRc flows is changed to a direction away from the nozzle N (that is, the Z1 direction). Therefore, as shown in FIG. 15, it is difficult for both the stream FRb caused by the minute vibration operation and the stream FRc caused by the circulation operation to reach a region downstream of the nozzle N in the nozzle flow path Nf. Therefore, the thickened ink Bu may continue to stay at the region.

(139) In addition, in the comparative embodiment, the stream FRb caused by the minute vibration operation eliminates thickening at the position inside the nozzle N and at the position closer to the side to which the Z1 direction extends than the nozzle N is in FIG. 13. However, the thickening may not be eliminated although depending on the degree of progress of thickening. The reason why

the thickening is not eliminated is that, when thickening in the nozzle N is progressed, there is an increase in flow path resistance due to the thickening over an area from the pressure chamber Ca to the nozzle N and ink may almost not flow even when the minute vibration operation is performed. When the piezoelectric elements PZ are driven without ink flowing, heat is accumulated and a load is applied to a circuit such as the drive circuit **14i**.

(140) Therefore, in the case of the liquid ejecting apparatus **100** according to the first embodiment, the circulation operation is started first and then the minute vibration operation is started. A series of operations in the liquid ejecting apparatus **100** according to the first embodiment will be described with reference to FIGS. **16** to **20**.

(141) 1-8. Operation in Liquid Ejecting Apparatus **100**

(142) FIG. **16** is a diagram showing a series of operations in the liquid ejecting apparatus **100**. While the series of operations shown in FIG. **16** is being performed, the liquid ejecting apparatus **100** may enter a printing preparation state which is a state where preparation for a printing process is performed, a state where the printing process is performed, and a printing-ended state where the printing process is finished. As shown in FIG. **16**, throughout a period from a time **T0** to a time **Td**, the state of the liquid ejecting apparatus **100** is the printing preparation state. Throughout a period from the time **Td** to a time **Te**, the liquid ejecting apparatus **100** performs the printing process. Throughout a period from the time **Te** to a time **Th**, the state of the liquid ejecting apparatus **100** is the printing-ended state.

(143) As shown in FIG. **16**, when a power source is turned on at the time **T0** in response to an operation performed by a user of the liquid ejecting apparatus **100** and the printing data **Img** is supplied from the host computer, the circulation control section **127** starts the circulation operation at a time **Ta**. Hereinafter, the user of the liquid ejecting apparatus **100** will be simply referred to as a “user”. A period **T0a** from the time **T0** to the time **Ta** is, for example, 5 seconds. Since the state of the vicinity of a nozzle at the time **T0a** is the same as a state shown in FIG. **12**, illustration and description thereof will be omitted. As shown in FIG. **16**, the circulation control section **127** continues, until the time **Th** is reached, the circulation operation which is started at the time **Ta**. The time **Ta** is an example of a “first time”. The time **Th** is an example of a “fourth time”.

(144) FIG. **17** is a view showing how the vicinity of the nozzle N is at the time **Ta**. As shown in FIG. **17**, when the circulation operation is started, the stream **FRc** along the nozzle flow path **Nf** is generated. FIG. **18** is a view showing how the vicinity of the nozzle N is in a period **Tab**. Ink staying in the nozzle flow path **Nf**, which is a portion of the thickened ink **Bu** staying inside and near the nozzle N, is discharged to the second common liquid chamber **R2** because of the stream **FRc**. The thickened ink **Bu** inside the nozzle N continues to stay throughout the period **Tab**.

(145) As shown in FIG. **16**, the minute vibration control section **123** starts the minute vibration operation at a time **Tb** later than the time **Ta**. The period **Tab** from the time **Ta** to the time **Tb** is shorter than the period **T0a** and is, for example, 0.5 seconds. The time **Tb** is an example of a “second time”. The period **Tab** is an example of “a time interval between the first time and the second time”.

(146) The minute vibration control section **123** continues, until a time **Tg** is reached, the minute vibration operation started at the time **Tb** and ends the minute vibration operation at the time **Tg**. The time **Tg** is earlier than the time **Th**. The time **Tg** is an example of a “fifth time”.

(147) As shown in FIG. **16**, the time **Tb** is earlier than the time **Th**. Therefore, the circulation control section **127** is continuing the circulation operation at the time **Tb**.

(148) FIG. **19** is a view showing how the vicinity of the nozzle N is at the time **Tb**. As shown in FIG. **19**, when the minute vibration operation is started, the stream **FRb** along the Z-axis is generated in the nozzle N. As shown in FIG. **19**, since the stream **FRb** reaches the nozzle flow path **Nf**, the stream **FRc** meanders because of the stream **FRb**. Specifically, at the position upstream of the vicinity of the nozzle N, a direction in which the stream **FRc** flows is changed to the direction away from the nozzle N (that is, the **Z1** direction).

(149) As shown in FIG. 16, the cap control section 125 starts the cap unsealing operation at a time T_c later than the time T_b . A period T_{bc} from the time T_b to the time T_c is shorter than the period T_{0a} and longer than the period T_{ab} . The period T_{bc} is, for example, 3 seconds. The cap control section 125 continues, until a time T_f is reached, the cap unsealing operation started at the time T_c and ends the cap unsealing operation at the time T_f . The time T_f is earlier than the time T_g . The time T_c is an example of a “third time”. The time T_f is an example of a “sixth time”. The period T_{bc} is an example of “a time interval between the second time and the third time”.

(150) As shown in FIG. 16, the time T_c is earlier than the time T_g and the time T_h . Therefore, at the time T_c , the circulation control section 127 is continuing the circulation operation, and the minute vibration control section 123 is continuing the minute vibration operation.

(151) FIG. 20 is a view showing how the vicinity of the nozzle N is in the period T_{bc} . As shown in FIG. 20, the stream FR_b eliminates thickening inside the nozzle N. In the first embodiment as well, similarly to the comparative embodiment, it is difficult for both the stream FR_b caused by the minute vibration operation and the stream FR_c caused by the circulation operation to reach a region downstream of the nozzle N in the nozzle flow path N_f . However, in the first embodiment, the thickened ink B_u staying at the region downstream of the nozzle N in the nozzle flow path N_f is eliminated in the period T_{ab} , and the thickening after the time T_b is eliminated through the minute vibration operation and the circulation operation. Therefore, local thickening as in the comparative embodiment is less likely to occur and the ejection property, which is one or both of the ejection amount and the ejection rate, can be maintained. Since the ejection property can be maintained, ejection failure can be suppressed. Examples of the ejection failure include a so-called dot omission in which ink is not ejected from the nozzle N, a flying curve phenomenon, and the like.

(152) Both the circulation operation and the minute vibration operation are operations of applying pressure to ink. However, as can be understood from FIG. 20, the minute vibration operation generates the stream FR_b along a direction in which the nozzle N extends (that is, the stream FR_b along the Z-axis) while the circulation operation generates the stream FR_c along the nozzle flow path N_f . The reason why the behavior differs between the circulation operation and the minute vibration operation will be described below.

(153) The circulation operation only generates an ink stream in the liquid ejection head 10 that flows from an upstream area and a downstream area in the liquid ejection head 10 and cannot generate a local stream flowing from the individual flow path PJ to the nozzle N. Meanwhile, regarding the minute vibration operation, the individual flow path PJ is in a state of being like a closed space since the narrowed portions Ap are present at both ends of the individual flow path PJ. Therefore, the ratio of a pressure transmitted to the nozzle N to a pressure applied to ink because of the minute vibration operation is larger than the ratio of a pressure transmitted to the narrowed portions Ap to the pressure applied to the ink because of the minute vibration operation.

Accordingly, a large pressure is transmitted to the nozzle N in comparison with an embodiment in which no narrowed portion Ap is present. Therefore, it is conceivable that the minute vibration operation according to the first embodiment generates the stream FR_b along the Z-axis that is large in comparison with the embodiment in which no narrowed portion Ap is present. In addition, in the first embodiment, the substantially same drive signals Com are supplied to the piezoelectric element PZ_a positioned upstream of the nozzle N and the piezoelectric element PZ_b positioned downstream of the nozzle N in one ejector D when the minute vibration operation is performed. Therefore, a pressure applied to ink from the piezoelectric element PZ_a and a pressure applied to the ink from the piezoelectric element PZ_b are merged with each other in the vicinity of the nozzle N which is positioned at the approximately center of the individual flow path PJ. It is conceivable that a portion of the merged pressure proceeds toward the nozzle N. Therefore, it is also conceivable that the minute vibration operation generates the stream FR_b along the Z-axis.

(154) In addition, in the case of the circulation operation, the stream FR_c along the nozzle flow path N_f is generated by a pressure difference between the upstream area and the downstream area

in the liquid ejection head **10**. However, since a pressure at a meniscus position and the atmospheric pressure are balanced, almost no stream along the Z-axis is generated. The meniscus is a surface of ink that is formed in the nozzle N. The pressure at the meniscus position is a pressure applied to the meniscus by the ink positioned inside the meniscus. Meanwhile, in the case of the minute vibration operation, there is a difference between the pressure at the meniscus position and the atmospheric pressure. Therefore, the stream FRb along the Z-axis can be generated.

(155) In addition, since the circulation operation generates a unidirectional stream flowing from the upstream area to the downstream area in the liquid ejection head **10**, the circulation operation has an effect of replacing the ink in the individual flow path PJ. Meanwhile, since the minute vibration operation generates a bidirectional stream that alternately repeats flowing in the Z1 direction and flowing in the Z2 direction, the minute vibration operation has an effect of stirring ink inside and near the nozzle N and an effect of diffusing ink thickening.

(156) As shown in FIG. **16**, the ejection control section **122** starts the printing ejection operation at the time Td later than the time Tc. The ejection control section **122** continues the printing ejection operation until the time Te later than the time Td is reached, and ends the printing ejection operation at the time Te. The liquid ejecting apparatus **100** in a period Tde from the time Td to the time Te will be described with reference to FIGS. **21** and **22**.

(157) FIG. **21** is a timing chart for description of the operation of the liquid ejecting apparatus **100** in the period Tde. The period Tde includes one or more unit periods Tu. As shown in FIG. **21**, the control module **120** outputs the latch signal LAT having pulses PlsL. Accordingly, the control module **120** defines the unit period Tu as a period from a rise of the pulse PlsL to the next rise of the pulse PlsL.

(158) The designation signal SI includes, with respect to one head main body **14**, individual designation signals Sd[1] to Sd[M] that designate the way in which the ejectors D[1] to D[M] are driven in each of the unit periods Tu. In addition, as shown in FIG. **21**, the control module **120** supplies, to the coupling state designation circuit **142**, the designation signal SI including the individual designation signals Sd[1] to Sd[M] in synchronization with a clock signal CL for one unit period Tu or each of a plurality of the unit periods Tu before the unit period Tu is started. In this case, the coupling state designation circuit **142** generates the coupling state designation signals SL1a[m], SL2a[m], SL1b[m], and SL2b[m] based on an individual designation signal Sd[m] in the unit period Tu, with respect to any m ranging from 1 to M.

(159) Note that, the individual designation signal Sd[m] according to the first embodiment is a signal for designation of any one of the printing ejection operation and the minute vibration operation with respect to the ejector D[m] in each unit period Tu, with respect to any m ranging from 1 to M. In the first embodiment, for example, it will be assumed that the individual designation signal Sd[m] is a 1-bit digital signal as shown in FIG. **22**, with respect to any m ranging from 1 to M.

(160) As shown in FIG. **21**, the drive signal generation circuit **190** outputs the drive signal Com-A having an ejection waveform PX that causes ink to be ejected from the nozzle N and the drive signal Com-B having a minute vibration waveform PS that causes ink in the nozzle N to vibrate to such a degree that no ink is ejected from the nozzle N. As can be understood from FIG. **21**, the ejection waveform PX and the minute vibration waveform PS are different from each other. The minute vibration waveform PS is an example of a “first waveform”. The ejection waveform PX is an example of a “second waveform”.

(161) Potentials at the start and end of the ejection waveform PX and potentials at the start and end of the minute vibration waveform PS are set to a reference potential V0. A manufacturer of the liquid ejecting apparatus **100** determines the ejection waveform PX and the minute vibration waveform PS such that a potential difference between a maximum potential VHx and a minimum potential VLx of the ejection waveform PX is greater than a potential difference between a minimum potential VLS of the minute vibration waveform PS and the reference potential V0.

Furthermore, the manufacturer of the liquid ejecting apparatus **100** determines the minute vibration waveform PS such that an ink stream in the Z1 direction that is generated in the nozzle N reaches the individual flow path PJ (more specifically, the nozzle flow path Nf) when the drive signal Com-B having the minute vibration waveform PS is supplied to the piezoelectric elements PZ. Note that, the ink stream in the Z1 direction is an example of “a liquid stream toward an individual flow path”.

(162) When the individual designation signal Sd[m] is for designation of the printing ejection operation with respect to the ejector D[m], the coupling state designation circuit **142** sets the coupling state designation signals SL1a[m] and SL1b[m] to a high level in the unit period Tu and sets the coupling state designation signals SL2a[m] and SL2b[m] to a low level in the unit period Tu, with respect to any m ranging from 1 to M. Accordingly, the ejector D[m] ejects ink in the unit period Tu so that a dot is formed on the medium PP.

(163) In addition, when the individual designation signal Sd[m] is for designation of the minute vibration operation with respect to the ejector D[m], the coupling state designation circuit **142** sets the coupling state designation signals SL1a[m] and SL1b[m] to a low level in the unit period Tu and sets the coupling state designation signals SL2a[m] and SL2b[m] to a high level in the unit period Tu, with respect to any m ranging from 1 to M. Accordingly, in the ejector D[m], ink in the nozzle N is vibrated in the unit period Tu and no dot is formed on the medium PP.

(164) The drive control section **121** generates, based on the printing data Img, an individual designation signal Sd for designation of any one of the printing ejection operation and the minute vibration operation with respect to each of the M ejectors D of each of the six head main bodies **14**. Therefore, in the period Tde, the minute vibration operation is temporarily stopped in the ejector D for which the printing ejection operation is designated, and the printing ejection operation is temporarily stopped in the ejector D for which the minute vibration operation is designated. For example, in one unit period Tu within the period Tde, the minute vibration operation is temporarily stopped in an ejector D[m3] for which the printing ejection operation is designated and the minute vibration operation is continued in an ejector D[m4] for which the minute vibration operation is designated. Furthermore, when the minute vibration operation is designated for the ejector D[m3] in the next unit period Tu after the above-described unit period Tu, the minute vibration operation is restarted and the printing ejection operation is temporarily stopped in the ejector D[m3]. Here, m3 and m4 are integers in a range from 1 to M and are integers different from each other.

(165) Although description about the period Tde has been made with reference to FIG. **21**, even when the minute vibration operation is performed in the period Tbc and a period Tcd, the drive control section **121** generates the individual designation signal Sd[m] for designation of the minute vibration operation, with respect to every m ranging from 1 to M. Accordingly, the minute vibration operation is performed in the period Tbc and the period Tcd as well. Note that, since the printing ejection operation is not performed in the period Tbc and the period Tcd, the drive control section **121** may generate the waveform designation signal dCom such that the drive signal Com-A set to have the reference potential Vbs and the drive signal Com-B having the minute vibration waveform PS are supplied to the head main bodies **14**.

(166) FIG. **22** is a description diagram for description of the coupling state designation signals SL1a[m], SL2a[m], SL1b[m], and SL2b[m], with respect to any m ranging from 1 to M. The coupling state designation circuit **142** decodes the individual designation signal Sd[m] according to FIG. **22** and generates the coupling state designation signals SL1a[m], SL2a[m], SL1b[m], and SL2b[m].

(167) As shown in FIG. **22**, the individual designation signal Sd[m] according to the first embodiment shows any of a value (1) for designation of the printing ejection operation and a value (0) for designation of the minute vibration operation. In addition, when the individual designation signal Sd[m] shows (1), the coupling state designation circuit **142** sets the coupling state designation signals SL1a[m] and SL1b[m] to a high level in the unit period Tu and when the

individual designation signal $Sd[m]$ shows (0), the coupling state designation circuit **142** sets the coupling state designation signals $SL2a[m]$ and $SL2b[m]$ to a high level in the unit period Tu . Otherwise, the coupling state designation circuit **142** sets each signal to a low level.

(168) The description will be made referring again to FIG. **16**. When the printing ejection operation ends at the time Te , the cap control section **125** ends the cap unsealing operation and starts the cap sealing operation at the time Tf later than the time Te . The minute vibration control section **123** ends the minute vibration operation at the time Tg later than the time Tf . Even in a period Tef from the time Te to the time Tf and a period Tfg from the time Tf to the time Tg , the minute vibration control section **123** generates the individual designation signal $Sd[m]$ for designation of the minute vibration operation, with respect to every m ranging from 1 to M . Accordingly, the minute vibration operation is performed in the period Tef and the period Tfg as well.

(169) The circulation control section **127** ends the circulation operation at the time Th later than the time Tg . A period Tgh from the time Tg to the time Th is shorter than the period Tab . Note that, the period Tgh is an example of “a time interval between the fourth time and the fifth time”.

1-9. Summary of First Embodiment

(170) Hereinafter, the liquid ejecting apparatus **100** according to the first embodiment will be described while using any integer $m1$ ranging from 1 to m and $m2$ which is an integer different from the integer $m1$ in a range from 1 to M .

(171) The liquid ejecting apparatus **100** according to the first embodiment includes the liquid ejection head **10** and the control module **120**. The control module **120** functions as the circulation control section **127** and the minute vibration control section **123**. The liquid ejection head **10** includes the piezoelectric element $PZa[m1]$ that is driven in response to supply of the drive signal Com , the nozzle $N[m1]$ that ejects ink by means of a pressure that is applied when the piezoelectric element $PZa[m1]$ is driven, and the individual flow path $PJ[m1]$ that communicates with the nozzle $N[m1]$, through which ink is supplied to the nozzle $N[m1]$, and through which ink not ejected from the nozzle $N[m1]$ is discharged. The circulation control section **127** controls the circulation operation of circulating the ink in the individual flow path $PJ[m1]$. The minute vibration control section **123** supplies the drive signal $Com-B$ having the minute vibration waveform PS to the piezoelectric element $PZa[m1]$ so as to control the minute vibration operation of causing ink in the nozzle $N[m1]$ to vibrate to such a degree that the ink is not ejected from the nozzle $N[m1]$. The circulation control section **127** starts the circulation operation at the time Ta . The minute vibration control section **123** starts the minute vibration operation at the time Tb later than the time Ta .

(172) As shown in FIG. **12** to FIG. **15**, in the comparative embodiment, it is difficult for both the stream FRb caused by the minute vibration operation and the stream FRc caused by the circulation operation to reach a region downstream of the nozzle N in the nozzle flow path Nf . Therefore, the thickened ink Bu may continue to stay at the region. Meanwhile, according to the first embodiment, the stream FRc caused by the circulation operation reaches the region downstream of the nozzle N in the nozzle flow path Nf in the period Tab . Therefore, it is possible to restrain the thickened ink Bu from staying. Therefore, according to the first embodiment, it is possible to suppress a decrease in ejection property in comparison with the comparative embodiment.

(173) In addition, the liquid ejecting apparatus **100** includes the cap **182** that seals the nozzle surface FN provided with the nozzle $N[m1]$. The control module **120** also functions as the cap control section **125**. The cap control section **125** controls an unsealing operation of unsealing the nozzle surface FN sealed by the cap **182**. The cap control section **125** starts the cap unsealing operation at the time Tc later than the time Tb .

(174) When the nozzle surface FN is unsealed, evaporation of a solvent in ink at the nozzle $N[m1]$ becomes remarkable and thus thickening may be rapidly progressed inside and near the nozzle $N[m1]$. In the first embodiment, the circulation operation and the minute vibration operation are started before the time Tc at which thickening of ink is rapidly progressed. Therefore, thickening of ink inside and near the nozzle $N[m1]$ can be suppressed in comparison with an embodiment in

which the circulation operation and the minute vibration operation are started after the time T_c .

(175) In addition, the period T_{ab} is shorter than that of the period T_{bc} . In other words, a period in which only the circulation operation is performed before the printing ejection operation is shorter than a period in which only the circulation operation and the minute vibration operation are performed before the printing ejection operation.

(176) It is preferable that the cap sealing operation is in progress in the period T_{bc} and the circulation operation and the minute vibration operation are performed for a certain period so that thickening is eliminated as much as possible before the start of the cap unsealing operation.

Meanwhile, the period T_{ab} does not need to be longer than necessary because only to be performed in the period T_{ab} is elimination of thickening in the nozzle flow path N_f . If the period T_{ab} is set to be longer than necessary, a period from a time at which the power source is turned on to the time T_e (that is, a period for which a user waits for the end of the printing process) is made long, which is not preferable. Therefore, according to the first embodiment, thickening can be eliminated as much as possible before the start of the cap unsealing operation and a time for which the user waits for the end of the printing process can be shortened in comparison with an embodiment in which the period T_{ab} is longer than the period T_{bc} .

(177) In addition, the control module **120** also functions as the ejection control section **122**. The ejection control section **122** supplies the drive signal Com-A having the ejection waveform PX to the piezoelectric element PZa[m1] so as to control the printing ejection operation of causing ink to be ejected from the nozzle N[m1]. The ejection control section **122** starts the printing ejection operation after the time T_c .

(178) At the time T_c , the circulation operation and the minute vibration operation have been already started and thickening of ink inside and near the nozzle N[m1] has been suppressed. When the printing ejection operation is started after the time T_c , the occurrence of ejection failure can be suppressed.

(179) In addition, the circulation control section **127** ends the circulation operation at the time T_h later than the end of the printing ejection operation and the minute vibration control section **123** ends the minute vibration operation at the time T_g later than the end of the printing ejection operation.

(180) In other words, the circulation operation and the minute vibration operation are continued until the printing ejection operation is ended. Therefore, it is possible to suppress the progress of thickening during the printing ejection operation.

(181) In addition, the time T_h is later than the time T_g . In other words, the liquid ejecting apparatus **100** ends the circulation operation after the minute vibration operation is ended. As shown in FIGS. **19** and **20**, when the minute vibration operation and the circulation operation are in progress, a direction in which the stream FRC caused by the circulation operation proceeds is changed by the stream FRb caused by the minute vibration operation. Therefore, at a position downstream of the vicinity of the nozzle N, the stream FRC meanders because of the stream FRb. The stream FRC meandering may cause stagnation at the position downstream of the vicinity of the nozzle N. When the stagnation occurs, thickened ink may stay. In an embodiment in which the minute vibration operation is ended after the circulation operation is ended, ink stops to flow in the individual flow path PJ with the stream FRC meandering. Therefore, the thickened ink may stay at the position downstream of the vicinity of the nozzle N.

(182) Meanwhile, in the first embodiment, the circulation operation is ended after the minute vibration operation is ended and thus meandering of the stream FRC caused by the circulation operation is eliminated in the period T_{gh} . When the meandering is eliminated, the occurrence of stagnation can be suppressed and thus thickened ink staying at the position downstream of the vicinity of the nozzle N can be suppressed.

(183) In addition, the period T_{gh} is shorter than that of the period T_{ab} . In other words, a period in which only the circulation operation is performed after the printing ejection operation is shorter

than a period in which only the circulation operation is performed before the printing ejection operation.

(184) Before the printing ejection operation, it is preferable to eliminate thickening as much as possible. However, after the printing ejection operation, it is sufficient to cause the ink in the individual flow path PJ to flow by means of the stream FRc caused by the circulation operation performed to a certain degree. This is because even if the circulation operation is performed for a long time after the printing ejection operation, the occurrence of slight evaporation of ink during a period between the end of the circulation operation and the next printing ejection operation cannot be suppressed. The circulation operation performed for an excessively long time only consumes power. Therefore, according to the first embodiment, thickening can be eliminated as much as possible before the printing ejection operation and power consumption of the liquid ejecting apparatus **100** can be reduced in comparison with an embodiment in which the period Tgh is longer than the period Tab.

(185) Further, the circulation control section **127** ends the circulation operation at the time Th, the minute vibration control section **123** ends the minute vibration operation at the time Tg, and the cap control section **125** ends the cap unsealing operation at the time Tf earlier than the time Th and the time Tg. In other words, at a time when the cap unsealing operation in which thickening of ink inside and near the nozzle N is rapidly progressed is ended, the circulation operation and the minute vibration operation are in progress.

(186) According to the first embodiment, rapid progress of thickening of ink inside and near the nozzle N can be suppressed in comparison with an embodiment in which the circulation operation and the minute vibration operation are ended before the cap unsealing operation is ended.

(187) In addition, the ejection control section **122** ends the printing ejection operation before the time Tf.

(188) At the time Tf, the circulation operation and the minute vibration operation are in progress so that thickening of ink inside and near the nozzle N[m1] is suppressed. When the printing ejection operation is ended before the time Tf, the occurrence of ejection failure can be suppressed throughout a period from the start of the printing ejection operation and the end of the printing ejection operation.

(189) The circulation control section **127** is continuing the circulation operation at the time Tb.

(190) In an embodiment in which the circulation operation is ended at the time Tb at which the minute vibration operation is started, progress of thickening of ink inside and near the nozzle N starts after the end of the circulation operation. Therefore, thickened ink may stay at the region downstream of the nozzle N in the nozzle flow path Nf as with the comparative embodiment in which the minute vibration operation is started first and then the circulation operation is started. Meanwhile, according to the first embodiment, the circulation operation is in progress at the time Tb. Therefore, thickening of ink inside and near the nozzle N can be suppressed and thickened ink staying at the region downstream of the nozzle N in the nozzle flow path Nf can be suppressed.

(191) The minute vibration control section **123** supplies the drive signal Com-B having the minute vibration waveform PS to the piezoelectric element PZa[m1] so as to control the minute vibration operation such that the stream FRb of ink in the Z1 direction that is generated in the nozzle N[m1] reaches the individual flow path PJ[m1].

(192) According to the first embodiment, an effect of stirring ink inside and near the nozzle N and an effect of diffusing thickening of ink can be enhanced in comparison with an embodiment in which the stream FRb does not reach the nozzle flow path Nf. However, as described with reference to FIG. **15**, when the stream FRb reaches the nozzle flow path Nf, the stream FRc meanders because of the stream FRb. Therefore, according to the first embodiment, an effect of stirring ink inside and near the nozzle N and an effect of diffusing thickening of ink can be enhanced in comparison with the embodiment in which the stream FRb does not reach the nozzle flow path Nf and thickened ink Bu continuously staying at the region downstream of the nozzle N

in the nozzle flow path Nf can be suppressed in comparison with the comparative embodiment.

(193) In addition, the individual flow path PJ[m1] includes the pressure chamber Ca[m1] positioned upstream of the nozzle N[m1] and the pressure chamber Cb[m1] positioned downstream of the nozzle N[m1], the piezoelectric element PZa[m1] corresponds to the pressure chamber Ca[m1], the liquid ejection head **10** further includes the piezoelectric element PZb[m1] that corresponds to the pressure chamber Cb[m1] and that is driven in response to supply of the drive signal Com, and the minute vibration control section **123** controls the minute vibration operation by driving both of the piezoelectric element PZa[m1] and the piezoelectric element PZb[m1].

(194) As described above, a pressure applied to ink from the piezoelectric element PZa and a pressure applied to the ink from the piezoelectric element PZb are merged with each other in the vicinity of the nozzle N which is positioned at the center of the individual flow path PJ and a portion of the merged pressure proceeds in a direction toward the nozzle N. Therefore, according to the first embodiment, the stream FRb along the Z-axis can be generated.

(195) In addition, the individual flow path PJ[m1] further includes the narrowed portion Ap1[m1] that is positioned upstream of the pressure chamber Ca[m1] and of which the flow path resistance is higher than that of the pressure chamber Ca[m1] and the narrowed portion Ap2 [m1] that is positioned downstream of the pressure chamber Cb[m1] and of which the flow path resistance is higher than that of the pressure chamber Cb[m1].

(196) When a pressure is applied to the pressure chamber Ca[m1] because of the minute vibration operation, the pressure is transmitted to a position upstream of the pressure chamber Ca[m1] and a position downstream of the pressure chamber Ca[m1]. Here, the narrowed portion Ap1[m1] positioned upstream of the pressure chamber Ca[m1] has a higher flow path resistance than that of the pressure chamber Ca[m1]. Therefore, the ratio of a pressure transmitted to the position downstream of the pressure chamber Ca[m1] to the pressure applied to the pressure chamber Ca[m1] is larger than the ratio of a pressure transmitted to the narrowed portion Ap1 to the pressure applied to the pressure chamber Ca[m1]. Similarly, the ratio of a pressure transmitted to the position downstream of the pressure chamber Cb[m1] to the pressure applied to the pressure chamber Cb[m1] is larger than the ratio of a pressure transmitted to the narrowed portion Ap2 to the pressure applied to the pressure chamber Cb[m1]. Therefore, with the minute vibration operation according to the first embodiment, it is possible to generate the stream FRb along the Z-axis that is large in comparison with the embodiment in which no narrowed portion Ap is present. Since the larger stream FRb can be generated, with the liquid ejecting apparatus **100** according to the first embodiment, it is possible to stir the thickened ink Bu inside and near the nozzle N in a short period of time.

(197) The liquid ejection head **10** further includes the piezoelectric element PZa[m2] that is driven in response to supply of the drive signal Com, the nozzle N[m2] that ejects ink by means of a pressure that is applied when the piezoelectric element PZa[m2] is driven, the individual flow path PJ[m2] that communicates with the nozzle N[m2], through which ink is supplied to the nozzle N[m2], and through which ink not ejected from the nozzle N[m2] is discharged, the common supply flow path CF1 that is coupled to both of the individual flow path PJ[m1] and the individual flow path PJ[m2] and through which ink is supplied to the individual flow path PJ[m1] and the individual flow path PJ[m2], the common discharge flow path CF2 that is coupled to both of the individual flow path PJ[m1] and the individual flow path PJ[m2] and through which the ink is discharged from the individual flow path PJ[m1] and the individual flow path PJ[m2], and the bypass flow path BP that is coupled to the common supply flow path CF1 and the common discharge flow path CF2 and of which the flow path resistance is lower than those of the individual flow path PJ[m1] and the individual flow path PJ[m2], the liquid ejecting apparatus **100** further includes the pump **157** that is provided upstream of the liquid ejection head **10**, and the circulation control section **127** controls the circulation operation by driving the pump **157**.

(198) An ink stream in the bypass flow path BP that is caused by the circulation operation is larger

than the stream FRc of ink in each of the individual flow path PJ[m1] and the individual flow path PJ[m2] that is caused by the circulation operation. Therefore, the stream FRc in the individual flow path PJ according to the first embodiment is smaller than the stream FRc in the individual flow path PJ according to an embodiment in which the bypass flow path BP is not present. When the stream FRc of ink is small, the stream FRb caused by the minute vibration operation becomes relatively large. Therefore, a degree to which the stream FRc meanders is large and it is difficult for the stream FRc to reach the region downstream of the nozzle N in the nozzle flow path Nf. Therefore, in the first embodiment, the circulation operation is started before the minute vibration operation so that a situation where the stream FRb is not present (that is, a situation where the stream FRc does not meander) can be achieved. Since the stream FRc does not meander, the thickened ink Bu staying in the region downstream of the nozzle N in the nozzle flow path Nf can be discharged.

(199) Accordingly, in the case of the first embodiment, air bubbles of ink can be retrieved since the bypass flow path BP is provided. Furthermore, the thickened ink Bu staying in the region downstream of the nozzle N in the nozzle flow path Nf can be discharged since the circulation operation is started before the minute vibration operation.

(200) In addition, a pressure that is applied to ink in the individual flow path PJ per unit period because of the minute vibration operation is larger than a pressure that is applied to the ink in the individual flow path PJ per unit period because of the circulation operation.

(201) In comparison with an embodiment in which a pressure that is applied per unit period because of the minute vibration operation is smaller than a pressure that is applied per unit period because of the circulation operation, a degree to which the stream FRc caused by the circulation operation meanders because of the stream FRb caused by the minute vibration operation is large and it is difficult for the stream FRc to reach the region downstream of the nozzle N in the nozzle flow path Nf. Therefore, in the first embodiment, the circulation operation is started before the minute vibration operation so that a situation where the stream FRc does not meander can be achieved.

2. Second Embodiment

(202) A second embodiment is different from the first embodiment in that a flushing operation is performed immediately before the printing ejection operation. Hereinafter, the second embodiment will be described.

(203) FIG. 23 is a functional block diagram showing an example of the configuration of a liquid ejecting apparatus 100-A according to a second embodiment. The liquid ejecting apparatus 100-A is different from the liquid ejecting apparatus 100 in that the liquid ejecting apparatus 100-A includes a control module 120-A instead of the control module 120. The control module 120-A is different from the control module 120 in that the control module 120-A functions as a drive control section 121-A instead of functioning as the drive control section 121. The drive control section 121-A is different from the drive control section 121 in that the drive control section 121-A functions as a flushing control section 124 as well. The flushing control section 124 controls a flushing operation of forcibly ejecting, from the nozzle N, ink that does not directly contribute to image formation. Specifically, the flushing control section 124 performs the flushing operation by supplying, to the piezoelectric elements PZ, the drive signal Com having a flushing waveform for forcible ejection of ink from the nozzle N. The flushing waveform may be any waveform as long as ink is ejected. However, for example, the flushing waveform may be a waveform that causes the pressure chambers C to vibrate more than in the case of the ejection waveform PX so that the thickened ink can be ejected. The flushing waveform is different from the minute vibration waveform PS because the flushing waveform is a waveform for ejection of ink.

(204) Note that, in the second embodiment, in addition to the ejection waveform PX, the flushing waveform is also an example of the “second waveform”. Similarly, in addition to the printing ejection operation, the flushing operation is also an example of the “ejection operation of causing

liquid to be ejected from the first nozzle”.

(205) A series of operations in the liquid ejecting apparatus **100-A** will be described with reference to FIG. **24**.

(206) 2-1. Operation in Liquid Ejecting Apparatus **100-A**

(207) FIG. **24** is a diagram showing a series of operations of the liquid ejecting apparatus **100-A**. The series of operations in the liquid ejecting apparatus **100-A** is different from the series of operations in the liquid ejecting apparatus **100** in that the flushing operation is started at a time $Td1$ later than the time Tc , the flushing operation is ended at a time $Td2$ later than the time $Td1$, and the printing ejection operation is performed throughout a period from the time $Td2$ to the time Te . As shown in FIG. **24**, throughout a period from the time $T0$ to the time $Td2$, the state of the liquid ejecting apparatus **100-A** is the printing preparation state. Throughout a period from the time $Td2$ to a time Te , the liquid ejecting apparatus **100-A** performs the printing process. Throughout a period from the time Te to a time Th , the state of the liquid ejecting apparatus **100-A** is the printing-ended state.

(208) In the second embodiment, the description will be made on assumption that all of M ejectors D of one head main body **14** perform the flushing operation in a period $T12$ from the time $Td1$ to the time $Td2$. However, a configuration in which a portion of the M ejectors D performs the flushing operation and the remainder of the M ejectors D performs the minute vibration operation may also be adopted.

(209) Since the printing ejection operation is not performed in the period $T12$, the flushing control section **124** generates the waveform designation signal $dCom$ such that the drive signal $Com-A$ having the flushing waveform is supplied to the head main body **14**, for example. Then, the flushing control section **124** generates the individual designation signal $Sd[m]$ for designation of the flushing operation, with respect to every m ranging from 1 to M . In the period $T12$, the individual designation signal $Sd[m]$ may show any of a value (1) for designation of the flushing operation and a value (0) for designation of the minute vibration operation. When the drive signal $Com-A$ having the flushing waveform is supplied to the head main body **14** and the individual designation signal $Sd[m]$ shows the value (1), the flushing operation is performed. In the period $T12$, the minute vibration operation is temporarily stopped in the ejector D for which the flushing operation is designated.

3. Modification Example

(210) Each of the above-described embodiments can be modified in various manners. A specific embodiment of modification will be described below. Any two or more embodiments selected from the following examples can be appropriately combined with each other as long as there is no contradiction.

3-1. First Modification Example

(211) In each of the above-described embodiments, the period Tgh is shorter than the period Tab . However, the disclosure is not limited thereto. For example, the period Tgh may be longer than the period Tab . In other words, a period in which only the circulation operation is performed after the printing ejection operation may be longer than a period in which only the circulation operation is performed before the printing ejection operation.

(212) When the period before the printing ejection operation is long, a period for which a user waits for the end of the printing process is made long. Meanwhile, at a time after the printing ejection operation, an image has been already formed on the medium PP and thus the length of a period for which a user waits for the end of the printing process is not changed even if the period Tgh is made long. Therefore, according to the first modification example, a period for which the user waits for the end of the printing process can be shortened and thickening after the printing ejection operation can be eliminated more in comparison with an embodiment in which the period Tgh is shorter than the period Tab .

3-2. Second Modification Example

(213) In the first embodiment and the first modification example based on the first embodiment, the printing ejection operation is started after the cap unsealing operation is started. However, the cap unsealing operation and the printing ejection operation may not be performed. For example, when the printing ejection operation is not performed for a long period and thus there is a possibility that thickening of ink is excessively progressed, the liquid ejecting apparatus **100** may start the circulation operation, start the minute vibration operation thereafter, end the minute vibration operation after a certain period elapses with the cap sealing operation being performed, and end the circulation operation after the end of the minute vibration operation.

3-3. Third Modification Example

(214) In addition, in the second embodiment and the first modification example based on the second embodiment, the printing ejection operation is started after the flushing operation is performed. However, the printing ejection operation may not be performed. For example, when the printing ejection operation is not performed for a long period and thus there is a possibility that thickening of ink is excessively progressed, the liquid ejecting apparatus **100-A** may start the circulation operation, start the minute vibration operation thereafter, start the cap unsealing operation after the start of the minute vibration operation, start the cap sealing operation without the printing ejection operation after performing the flushing operation after the start of the cap unsealing operation, end the minute vibration operation after the start of the cap sealing operation, and end the circulation operation after the end of the minute vibration operation.

3-4. Fourth Modification Example

(215) In each of the above-described embodiments, the period T_{ab} is shorter than the period T_{bc} . However, the disclosure is not limited thereto. For example, the period T_{ab} may have the same length as the period T_{bc} and may be longer than the period T_{bc} .

3-5. Fifth Modification Example

(216) In each of the above-described embodiments, the time T_h is later than the time T_g . However, the disclosure is not limited thereto. For example, the time T_h may be the same time as the time T_g and may be earlier than the time T_g .

3-6. Sixth Modification Example

(217) In each of the above-described embodiments, one individual flow path PJ includes two pressure chambers C. However, one individual flow path PJ may include one pressure chamber C. Examples of an embodiment in which one individual flow path PJ includes one pressure chamber C include two embodiments as follows. In the first of the two embodiments, all of the M individual flow paths PJ include pressure chambers C_a without the pressure chambers C_b . In the second of the two embodiments, odd-numbered individual flow paths PJ of the M individual flow paths PJ include the pressure chambers C_a without the pressure chambers C_b and even-numbered individual flow paths PJ include the pressure chambers C_b without the pressure chambers C_a .

3-7. Seventh Modification Example

(218) In each of the above-described embodiments, the liquid ejection head **10** constitutes a line head. However, the disclosure is not limited to such a configuration and a serial type configuration in which the liquid ejection head **10** reciprocates along the X-axis may also be adopted.

3-8. Eighth Modification Example

(219) The liquid ejecting apparatus **100** described in each of the above-described embodiments can be adopted for various devices such as a facsimile machine and a copier in addition to a device dedicated to printing. However, the purpose of use of the liquid ejecting apparatus of the present disclosure is not limited to printing. For example, a liquid ejecting apparatus that ejects a solution of a coloring material is used as a manufacturing device forming a color filter of a liquid crystal display device. In addition, a liquid ejecting apparatus that ejects a solution of a conductive material is used as a manufacturing device forming a wire or an electrode on a wiring substrate.

Claims

1. A liquid ejecting apparatus comprising: a liquid ejection head including a first piezoelectric element that is driven in response to supply of a drive signal, a first nozzle that ejects liquid by a pressure that is applied when the first piezoelectric element is driven, and a first individual flow path that communicates with the first nozzle, through which the liquid is supplied to the first nozzle, and through which the liquid not ejected from the first nozzle is discharged; a circulation control section that controls a circulation operation of circulating the liquid in the first individual flow path; a minute vibration control section that supplies a drive signal having a first waveform to the first piezoelectric element so as to control a minute vibration operation of causing the liquid in the first nozzle to vibrate to such a degree that the liquid is not ejected from the first nozzle; and an ejection control section that supplies, to the first piezoelectric element, a drive signal having a second waveform different from the first waveform so as to control an ejection operation of causing the liquid to be ejected from the first nozzle, wherein the circulation control section starts the circulation operation at a first time and ends the circulation operation at a fourth time, the minute vibration control section starts the minute vibration operation at a second time later than the first time and ends the minute vibration operation at a fifth time, and the ejection control section starts the ejection operation at a seventh time later than the second time and earlier than the fifth time, and the ejection control section ends the ejection operation at an eighth time earlier than each of the fourth time and the fifth time.
2. The liquid ejecting apparatus according to claim 1, further comprising: a cap that seals a nozzle surface provided with the first nozzle; and a cap control section that controls an unsealing operation of unsealing the nozzle surface sealed by the cap, wherein the cap control section starts the unsealing operation at a third time later than the second time.
3. The liquid ejecting apparatus according to claim 2, wherein a time interval between the first time and the second time is shorter than a time interval between the second time and the third time.
4. The liquid ejecting apparatus according to claim 2, wherein the seventh time is later than the third time.
5. The liquid ejecting apparatus according to claim 1, wherein the fourth time is later than the fifth time.
6. The liquid ejecting apparatus according to claim 5, wherein a time interval between the fourth time and the fifth time is shorter than a time interval between the first time and the second time.
7. The liquid ejecting apparatus according to claim 5, wherein a time interval between the fourth time and the fifth time is longer than a time interval between the first time and the second time.
8. The liquid ejecting apparatus according to claim 1, further comprising: a cap that seals a nozzle surface provided with the first nozzle; and a cap control section that controls an unsealing operation of unsealing the nozzle surface sealed by the cap, wherein the circulation control section ends the circulation operation at the fourth time, and the cap control section ends the unsealing operation at a sixth time earlier than the fourth time and the fifth time.
9. The liquid ejecting apparatus according to claim 8, wherein the ejection control section ends the ejection operation at the eighth time earlier than the sixth time.
10. The liquid ejecting apparatus according to claim 1, wherein the circulation control section is continuing the circulation operation at the second time.
11. The liquid ejecting apparatus according to claim 1, wherein the minute vibration control section supplies the drive signal having the first waveform to the first piezoelectric element so as to control the minute vibration operation such that a liquid stream toward the first individual flow path reaches the first individual flow path, the liquid stream being generated in the first nozzle.
12. The liquid ejecting apparatus according to claim 1, wherein the first individual flow path includes a first pressure chamber positioned upstream of the first nozzle and a second pressure

chamber positioned downstream of the first nozzle, the first piezoelectric element corresponds to the first pressure chamber, the liquid ejection head further includes a second piezoelectric element that corresponds to the second pressure chamber and that is driven in response to supply of a drive signal, and the minute vibration control section controls the minute vibration operation by driving both of the first piezoelectric element and the second piezoelectric element.

13. The liquid ejecting apparatus according to claim 12, wherein the first individual flow path further includes a first narrowed portion that is positioned upstream of the first pressure chamber and of which a flow path resistance is higher than that of the first pressure chamber and a second narrowed portion that is positioned downstream of the second pressure chamber and of which a flow path resistance is higher than that of the second pressure chamber.

14. The liquid ejecting apparatus according to claim 1, wherein the liquid ejection head further includes a third piezoelectric element that is driven in response to supply of a drive signal, a second nozzle that ejects the liquid by a pressure that is applied when the third piezoelectric element is driven, a second individual flow path that communicates with the second nozzle, through which the liquid is supplied to the second nozzle, and through which the liquid not ejected from the second nozzle is discharged, a common supply flow path that is coupled to both of the first individual flow path and the second individual flow path and through which the liquid is supplied to the first individual flow path and the second individual flow path, a common discharge flow path that is coupled to both of the first individual flow path and the second individual flow path and through which the liquid is discharged from the first individual flow path and the second individual flow path, and a bypass flow path that is coupled to the common supply flow path and the common discharge flow path and of which a flow path resistance is lower than those of the first individual flow path and the second individual flow path, the liquid ejecting apparatus further includes one or more flow mechanisms provided at one or both of a position upstream of the liquid ejection head and a position downstream of the liquid ejection head, and the circulation control section controls the circulation operation by driving the one or more flow mechanisms.

15. The liquid ejecting apparatus according to claim 14, wherein a pressure that is applied to the liquid in the first individual flow path per unit period because of the minute vibration operation is larger than a pressure that is applied to the liquid in the first individual flow path per unit period because of the circulation operation.

16. A liquid ejecting apparatus comprising: a liquid ejection head including a first piezoelectric element that is driven in response to supply of a drive signal, a first nozzle that ejects liquid by a pressure that is applied when the first piezoelectric element is driven, and a first individual flow path that communicates with the first nozzle, through which the liquid is supplied to the first nozzle, and through which the liquid not ejected from the first nozzle is discharged; a circulation control section that controls a circulation operation of circulating the liquid in the first individual flow path; and a minute vibration control section that supplies a drive signal having a first waveform to the first piezoelectric element so as to control a minute vibration operation of causing the liquid in the first nozzle to vibrate to such a degree that the liquid is not ejected from the first nozzle, wherein the circulation control section starts the circulation operation at a first time, the minute vibration control section starts the minute vibration operation at a second time later than the first time, the first individual flow path includes a first pressure chamber positioned upstream of the first nozzle and a second pressure chamber positioned downstream of the first nozzle, the first piezoelectric element corresponds to the first pressure chamber, the liquid ejection head further includes a second piezoelectric element that corresponds to the second pressure chamber and that is driven in response to supply of a drive signal, and the minute vibration control section controls the minute vibration operation by driving both of the first piezoelectric element and the second piezoelectric element.

17. A liquid ejecting apparatus comprising: a liquid ejection head including a first piezoelectric element that is driven in response to supply of a drive signal, a first nozzle that ejects liquid by a

pressure that is applied when the first piezoelectric element is driven, and a first individual flow path that communicates with the first nozzle, through which the liquid is supplied to the first nozzle, and through which the liquid not ejected from the first nozzle is discharged; a circulation control section that controls a circulation operation of circulating the liquid in the first individual flow path; and a minute vibration control section that supplies a drive signal having a first waveform to the first piezoelectric element so as to control a minute vibration operation of causing the liquid in the first nozzle to vibrate to such a degree that the liquid is not ejected from the first nozzle, wherein the circulation control section starts the circulation operation at a first time, the minute vibration control section starts the minute vibration operation at a second time later than the first time, the liquid ejection head further includes: a third piezoelectric element that is driven in response to supply of a drive signal; a second nozzle that ejects the liquid by a pressure that is applied when the third piezoelectric element is driven; a second individual flow path that communicates with the second nozzle, through which the liquid is supplied to the second nozzle, and through which the liquid not ejected from the second nozzle is discharged; a common supply flow path that is coupled to both of the first individual flow path and the second individual flow path and through which the liquid is supplied to the first individual flow path and the second individual flow path; a common discharge flow path that is coupled to both of the first individual flow path and the second individual flow path and through which the liquid is discharged from the first individual flow path and the second individual flow path; and a bypass flow path that is coupled to the common supply flow path and the common discharge flow path and of which a flow path resistance is lower than those of the first individual flow path and the second individual flow path, the liquid ejecting apparatus further comprises one or more flow mechanisms provided at one or both of a position upstream of the liquid ejection head and a position downstream of the liquid ejection head, the circulation control section controls the circulation operation by driving the one or more flow mechanisms, and a pressure that is applied to the liquid in the first individual flow path per unit period because of the minute vibration operation is larger than a pressure that is applied to the liquid in the first individual flow path per unit period because of the circulation operation.
